Roboteam

HUMAN DRIVEN

ROBOX System Technical Data

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1. Introduction

The ROBOX System is a modular, containerized unmanned platform designed to enhance ground maneuver and force protection through autonomous and remotely operated assets. Each 20-ft control container houses two fully equipped unmanned ground vehicles (UGVs) and all supporting subsystems — including power generation, communications, and operator stations — forming a self-contained, deployable mission node.

The system provides a lethal operational capability that enables exposure and engagement of concealed threats through robotic stimulation–response operations, real-time detection of enemy forces and explosive traps, rapid neutralization, and precise direction of friendly elements.

ROBOX transforms the tactical landscape by expanding the security perimeter of maneuvering and border forces, increasing unit lethality, and minimizing soldier exposure.

This document defines the ROBOX technical baseline, architecture, and specifications (mechanical, electrical, software, and communication subsystems) for evaluation

2. System Overview

2.1. General Info

The ROBOX System is a modular, containerized mission mode that shifts from a transport unit into a protected command-and-control hub once deployed. Each 20-ft container carries two UGVs, operator consoles, power systems, communications equipment and an extendable mast for RF/cellular/GPS antennas.

The system offers versatile control modes — UGVs can be operated directly from the container or remotely via standalone ROCU controllers, allowing both centralized and distributed operations across the battlefield.

Workflow:

After deployment, the UGVs are unloaded for independent maneuver and mission execution. The container then becomes the stationary control center, providing power, data links, and long-range communication through its mast. From this protected node, operators manage surveillance, stimulation–response missions, threat exposure, and real-time engagement coordination.

2.2. System BOM

ROBOX
(ROBOX-V01-00001)

BISON UGV
(VHCR-V01-00001)
X2

UGV Controlled
Container
(RTCC-V01-00001)

Drone COM Set
(SET00135-A)

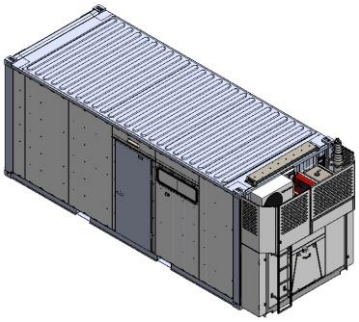
ROCU-10
(OEM00356)
X2

Operator Box
(ASSY800388-A)
X2

Tripod Relay Set
(SET00137-A)

System Overview

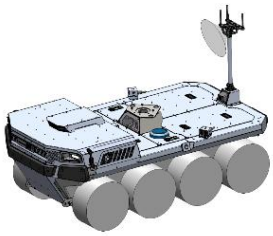
2.2.1. Overview



UGV Controlled Container



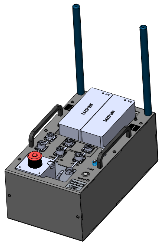
ROCU-10



BISON UGV



Drone Set



Operator Box

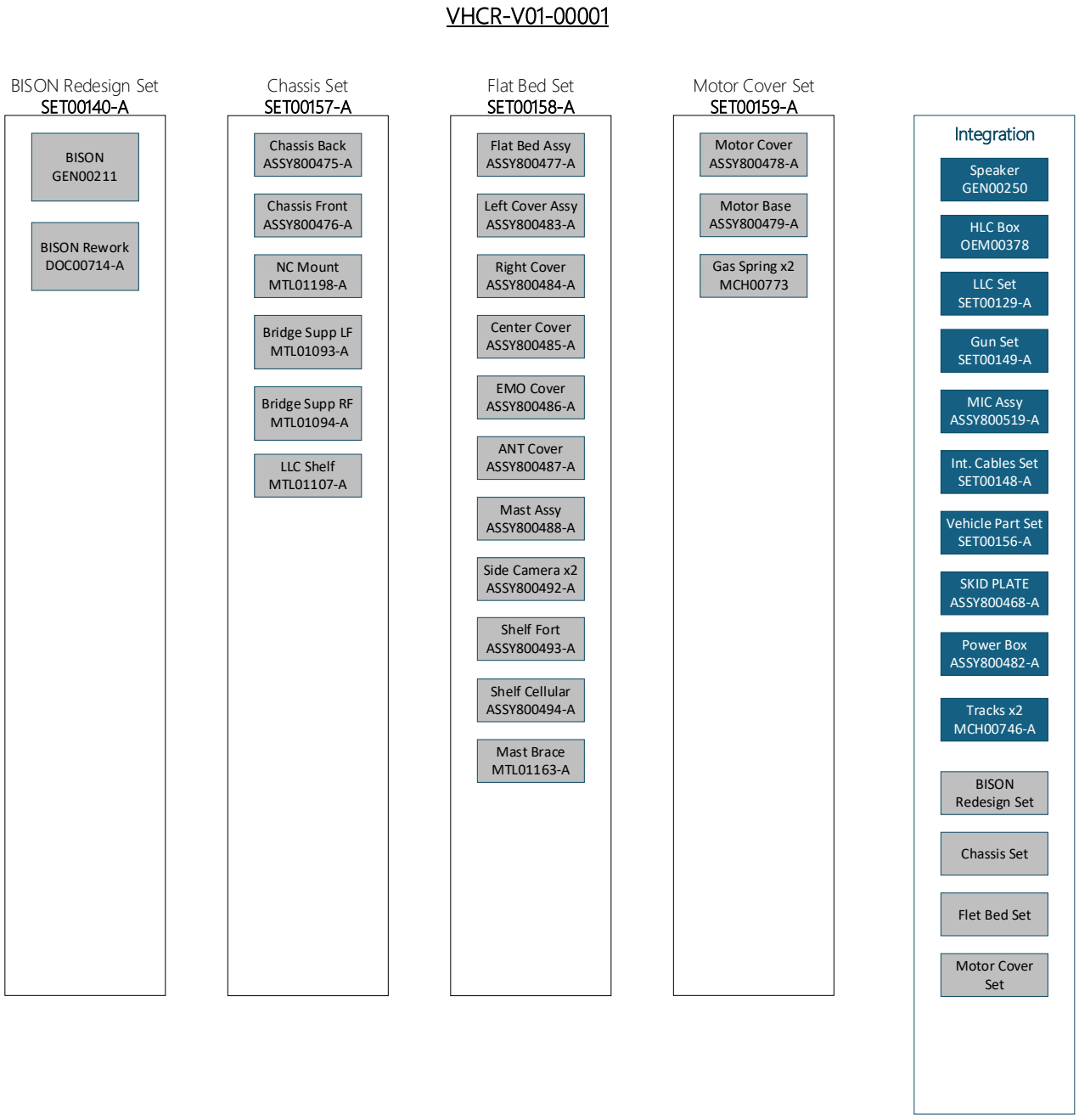


Tripod Relay Set

2.3. System High-Level Technical Specification

2.3.1. BISON UGV

2.3.1.1. Main BOM



2.3.1.2. Technical Specification

- ARGO Vehicle (Before control set assembly)
  - General Info

    Aurora is ARGO's premium XTV platform, equipped with a reliable VX 950 V-Twin engine.
  - Overview



  - Technical Specification

Parameter	Description
ENGINE	VX 950
CYLINDER	V-TWIN
FUEL SYSTEM	EFI
COOLING	FAN COOLED
ALTERNATOR OUTPUT (MAX)	50 AMPS
CHASSIS	8X8
SERVICE BRAKES	HYDRAULIC
STEERING BRAKES	HYDRAULIC
AUTO CHAIN LUBRICATION	ACCESSORY
EASY ACCESS GREASE SYSTEM (EAGS)	STANDARD
BRAKE COOLING FAN	ACCESSORY
TRANSMISSION	ADMIRAL W/APS

DRIVE BELT	STANDARD DUTY
FULL SKID PLATE	STANDARD
BRUSHGUARD	ACCESSORY
WINCH	ACCESSORY
FRONT RACK	ACCESSORY
CARGO BOX	ACCESSORY
BILGE PUMP	STANDARD
ROPS	ACCESSORY
ROOF	ACCESSORY W/ROPS
LOAD CAPACITY - LAND	1,080 lb (490 kg)
LOAD CAPACITY - WATER	880 lb (400 kg)
TOWING CAPACITY	1,800 lb (816 kg)
SEATING CAPACITY - LAND	6
SEATING CAPACITY - WATER	4
FUEL CAPACITY	8.5 gal (32 L)
SPEED - LAND	25 mph (40 Km/h)
SPEED - WATER	3 mph (5 Km/h)
WEIGHT	1,470 lb (668 kg)
LENGTH (includes winch when standard)	126 in (3,023 mm)
WIDTH	60 in (1,524 mm)
HEIGHT	51 in (1,295 mm)
AXLE BEARING EXTENSIONS	STANDARD
ENTRY STEP	ACCESSORY

TIRES	ARGO XT117 25X12-9
WHEEL	STEEL BEADLOCK
GROUND PRESSURE (18" RUBBER TRACKS)	0.88 psi (6.1 kPa)
GROUND CLEARANCE (TIRES)	10 in (254 mm)
GROUND CLEARANCE (TRACKS)	11 in (279 mm)

- Tracks (Optional)





Argo 18" Rubber tracks with no wheel spacers or lug extensions. The kit is designed for the Aluminum beadlock rims and new steel off-set rims that can be turned around for the proper offset when installing tracks. Kit includes all assembly hardware for installing tracks on the left and right sides of your Argo models listed below. The tracks add excellent snow flotation, and tire protection and are very quiet compared to the plastic tracks when running on hard-pack terrain. If your environment has mud or snow and you need a light footprint, these are the tracks you need for your Argo 8x8.

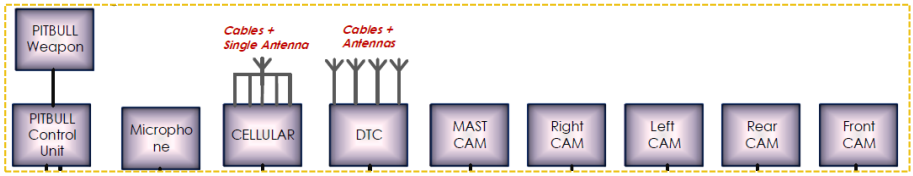
Adding these tracks to your Argo will make the overall width of the vehicle right at **71.5"**. This track kit will not work on any other model other than those listed below **or** can be used if your model already has wheel spacers and lug extenders installed and you just need replacement tracks

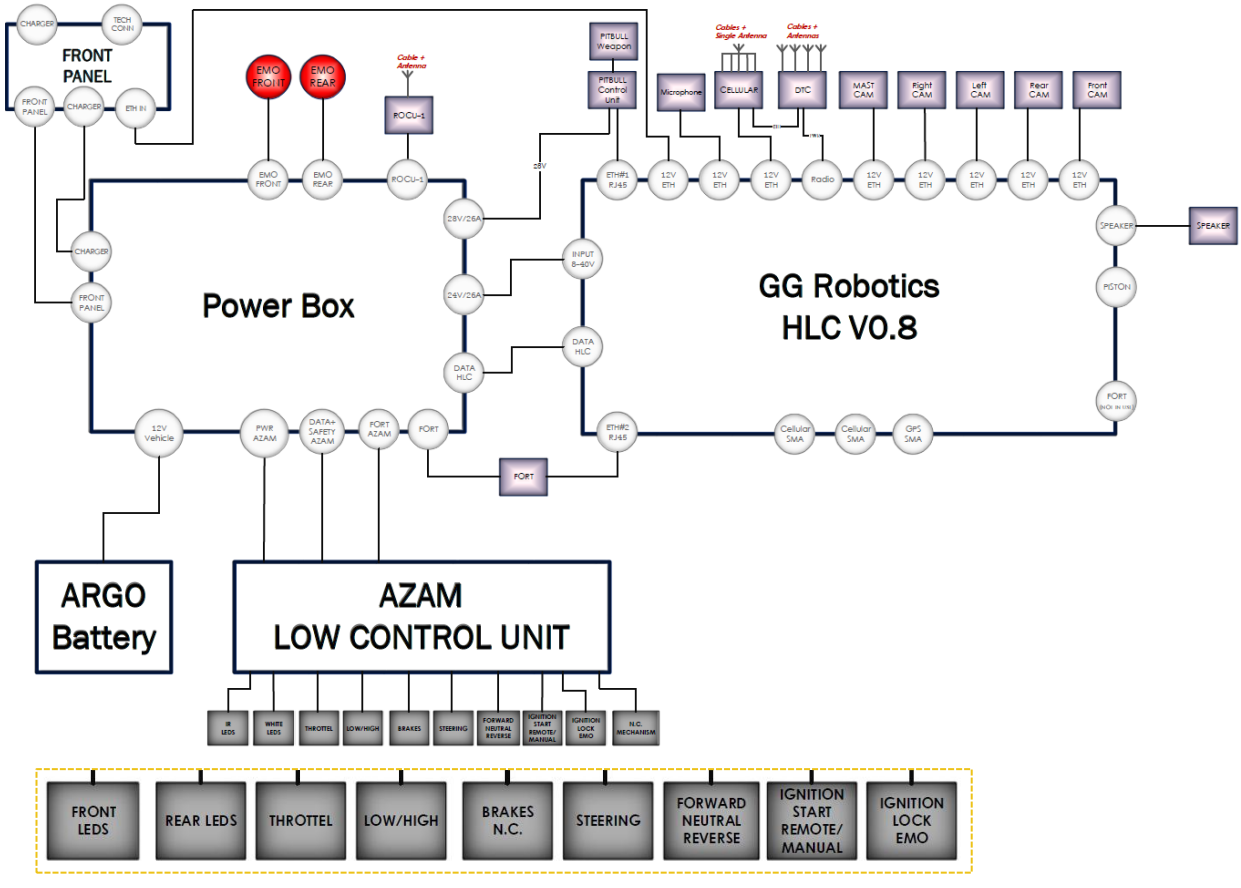
- Vehicle Bizon Configuration

Parameter	Description
ALTERNATOR OUTPUT (MAX)	55 AMPS @3000RPM, 15 AMP @1100RPM
WEIGHT	2094 lb (950 kg) (1100KG) WITH TRACKS
LENGTH (includes winch when standard)	118 in (3,000 mm)
LOAD CAPACITY - LAND	661 lb (300 kg)

- Vehicle Low and High Level Control System
  - General Info

    We take the vehicle and install the main structure with system control components.
    The low- and high-level control system controls all vehicle systems:
    Drive system
    Cameras (Day and Thermal)
    Communication
    Gun system
    LEDs (White)
    Speaker
    Microphone
    EMO
    ROCU-10 (Controller)
  - See main structure and component details in section 3.2
  - High-Level Architecture





- Cameras
  - General info

    The vehicle has four day&night (Thermal) cameras (front, left, right, back).
  - Overview



  - Technical Specification

Parameter	Description
Dimensions (LxWxH)	6.7x7.6x4.6 cm
Operational Temp	-40°C to 60°C
F.O.V	D=90° ,H=48°, V=38°
Resolution	640×5120 12µm 720P (MAX. 2688×1520)
Night View	Thermal
Environment	IP66

- MAST Camera
  - General info

    Mast camera placed at the back of the vehicle, looking forward for a deep view.
  - Overview



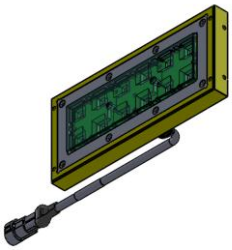
  - Technical Specification

Parameter	Description
Dimensions (LxWxH)	80x10.9x6 cm
Operational Temp	-25°C to 50°C
F.O.V	Wide Camera, D=140° ,H=105°, V=66°
Resolution	6 MP
Night View	IR Supported
Environment	IP67

- White LED
  - General Info

2X Automotive Floodlight LEDs for vehicle night-vision driving; assemble at the front of the vehicle.

  - Overview



  - Technical Specification

Parameter	Description
Power	12V, 24W
White Light Characteristics	5700-6000K; 85CRI
Cyan Light Characteristics	490-515nm
Viewing angle	60 deg
Operational Temp	-40°C to 85°C
Protection	IP67
Internal Potting	BLACK
Top Side Grinded	P 300

- Communication

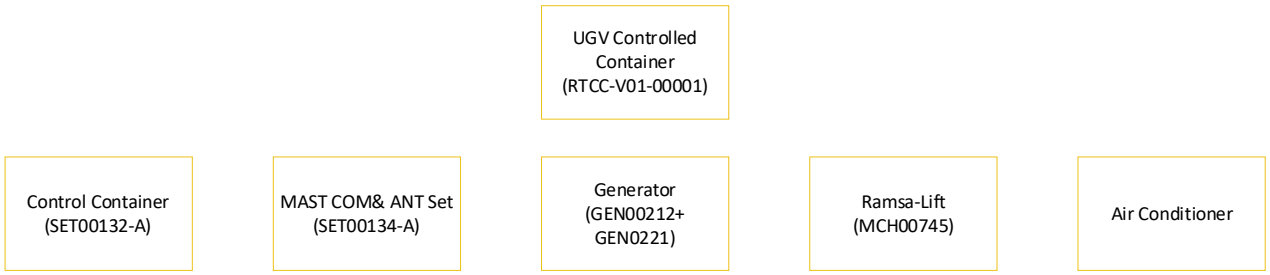
See section 3.5

- Gun System

See section 3.8

2.3.2. Tactical Operation Center (Controlled Container)

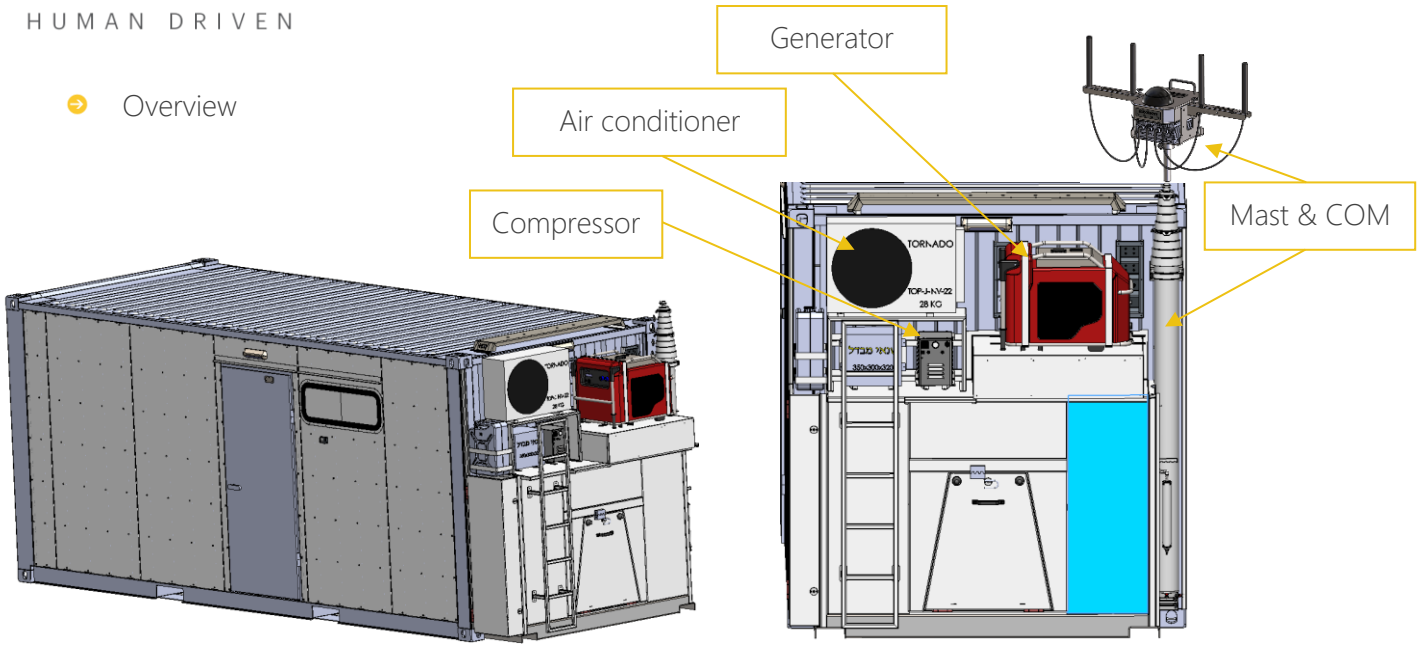
2.3.2.1. Main BOM



2.3.2.2. Technical Specification

- Tactical Operation Center (Controlled Container)
  - General Info
    - The **Tactical Operation Center** serves as the central element of the ROBOX system — a modular 20-ft ISO unit functioning both as a **transport platform** and as a **command-and-control center** once deployed in the field.
      During transport, the container carries **two UGVs**, their accessories, and mission payloads. Upon arrival, the UGVs are unloaded for independent operation, and the container transforms into a fully equipped control hub.
    - Inside the container are the **operator consoles**, communication systems, power generation and storage components, and an extendable **mast** for RF, cellular, and GPS antennas. This configuration provides long-range, secure connectivity to all deployed assets, including UGVs, relay units, and drone COM sets.
    - The container includes a **Generator**, and automatic power management that support sustained autonomous operation. Its **communication backbone** combines IP-mesh and tactical radios with multi-SIM cellular and GPS modules, enabling both direct and distributed control.
    - Built for field durability, the container is **climate-controlled, dust-protected**, and fully compliant with **MIL-STD-810G** environmental standards.
    - Once deployed, the container provides a secure, self-contained operational environment from which operators can conduct **surveillance, stimulation-response missions, remote engagements, and full mission coordination** with minimal setup time

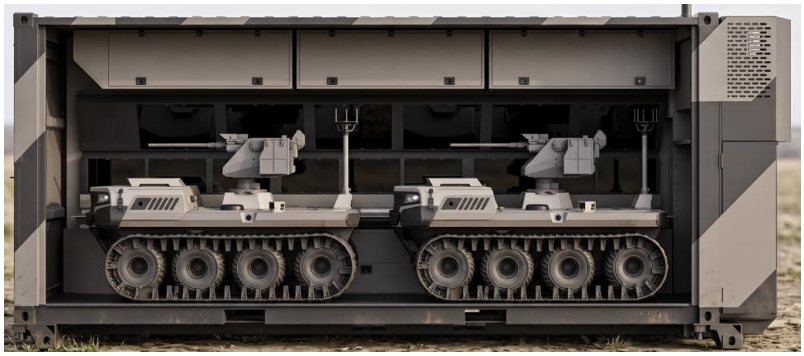
- Overview



- Technical Specification (Container)

Parameter	Description
Dimensions (LxWxH)	6.953x2.531x2.785 m
Length with RAMSA	Length=7.035m
Type	DVN (Steel), fit for truck transport
Internal Isolation	Structural Insulated Panels

- Container Layout-Transport (Inside container)



- Container Layout-Control (Mission)



MAST

General Info

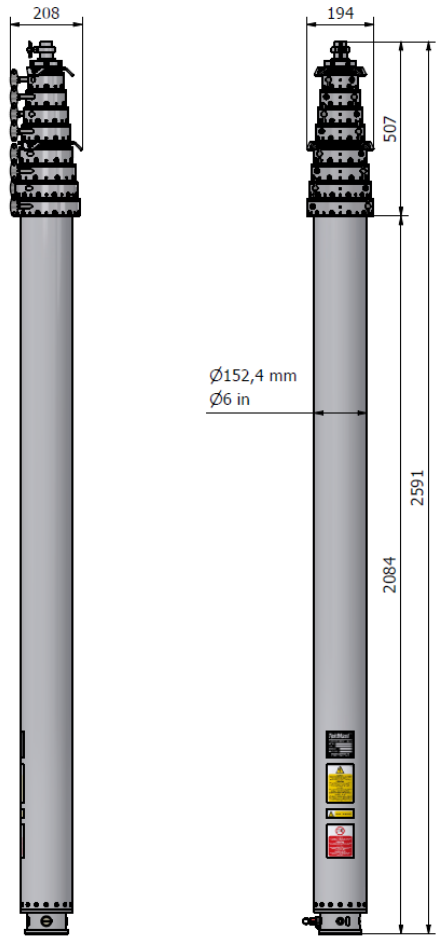
The mast supports a communication box (RF, Cellular & GPS) mounted on the container and connected to the operation box, allowing long-distance vehicle control (up to 10km for RF communication) from the container.

For the communication box, see section 2.3.8.

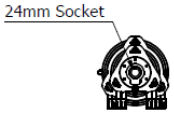


For communication architecture, see section 3.6.4

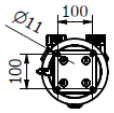
Overview



Top (1 : 10)



Bottom (1 : 10)



TekMast

Part Number: **54583-150-COFC-G**
Description: **VPT-6-150-9-NC Mast**

Standard features:

- Air operated telescopic mast with ***clamp collars***
- Seamless drawn aluminium tubes
- Equipped with overpressure / drain valve
- Top & middle guy collar
- Flat base flange
- 24mm top socket
- (Without internal cable)

Retracted height: 2591 mm
Extended height: 15065 mm
Wall thickness of mast sections: 3,25 mm
Base tube Ø: 6in
Top tube Ø: 2 in
Number of mast sections: 9
Max. headload: 35 kg - 0,4 m²
Max. wind speed guyed: 130 km/h - 80 mph
Max. wind speed unguyed: 60 km/h - 35 mph
(0,1 m² at 100 km/h unguyed)
Max. working pressure: 2,5 bar - 36 psi
Material of mast sections: Aluminium tube EN AW-6082-T6
Standard Finish: Natural anodised aluminium

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VPT-6-150-9-NC Mast	DRAWN BY: G. Philips		DATE: 17/03/2026
	WEIGHT: 66,5 kg		VERSION: 1.0
TEKSAM COMPANY NV WOUDSTRAAT 21, 3600 GENK, BELGIUM - WWW.TEKSAMCO.COM	SIZE: A3	SCALE: 1 : 10	DRAWING NO: TBC

Technical Specification

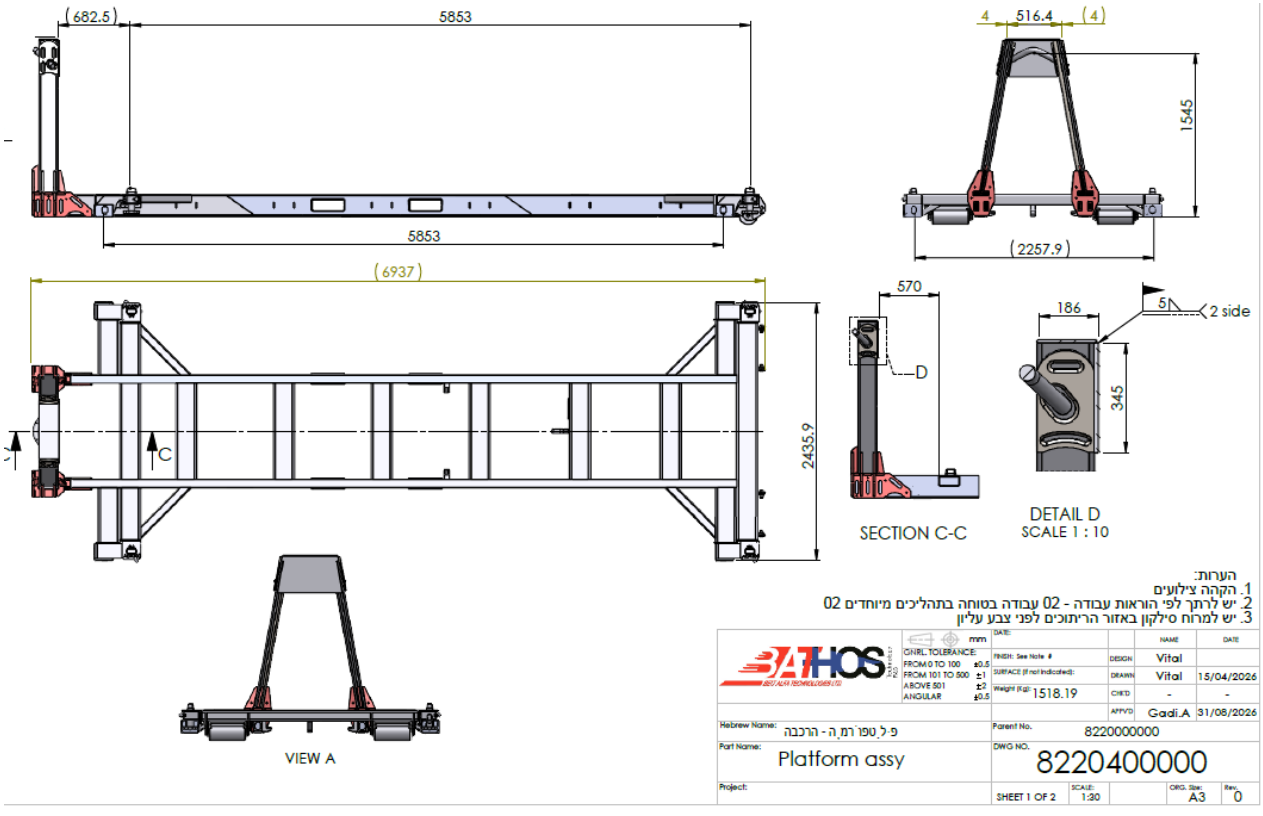
Parameter	Description
Height	5-15 m
Max Head Load	10 kg

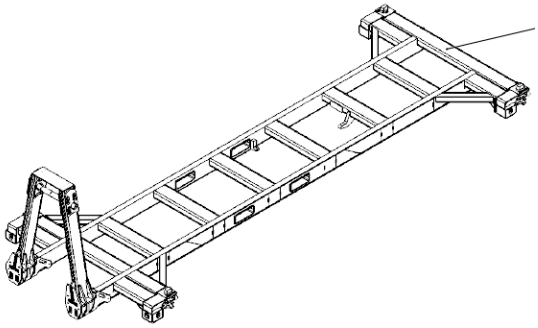
- RAMSA-Lift
  - General Info

    Container load/unload mount to track
  - Overview



  - Technical Specification (Dimensions in mm)





- Air conditioning
  - General Info

Electra Inverter AC



  - Technical Specification

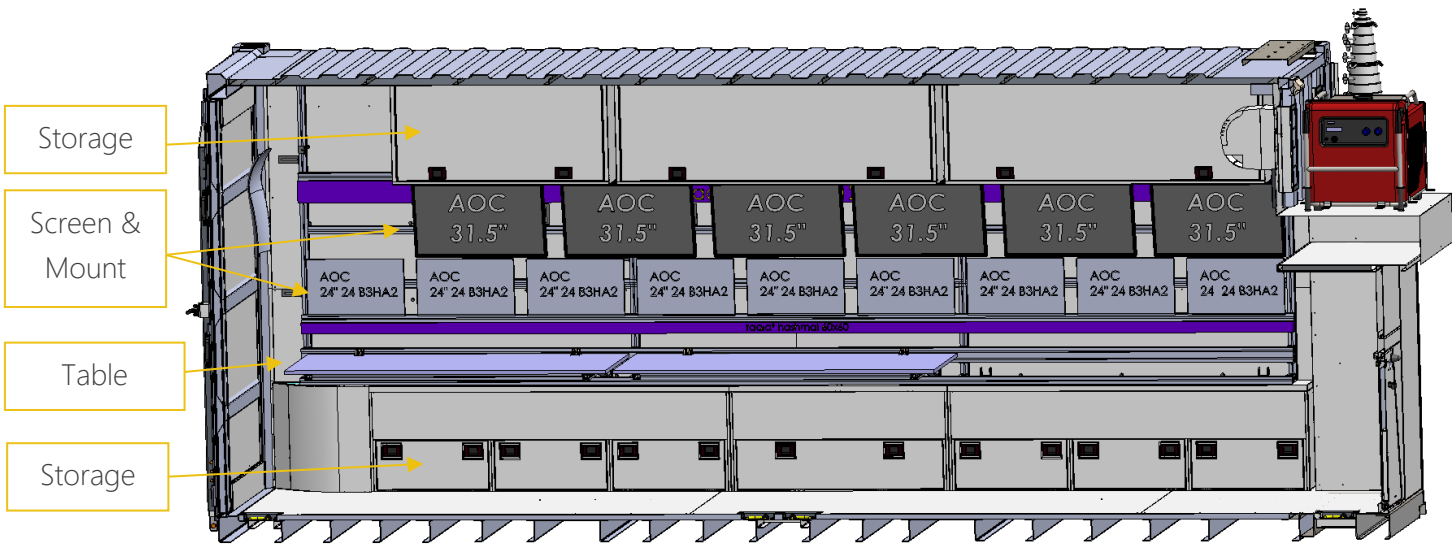
Parameter	Description
Model	Classic INV 170 / CMD12DCB
Fuse	10A (C)
Power Factor (cos φ)	0.92
IP Rating	IPX4
Revision	A
Pipe Length	8m < L < 20m
Refrigerant Charge	15 g/m
Rated Power	2300 W
Power Input (Pe)	1550 W
High Pressure (Pi/Po)	4.2 / 4.2 MPa
Low Pressure (Ps/Pd)	1.5 / 4.2 MPa
Rated Cooling	3.45 kW
Rated Heating	3.64 kW
Net Weight	20.6 kg

- Generator

See section 3.4.1.1 (Energy)

- Control Parts
  - General Info

    Inside the Tactical Operation Center we have up to nine operation stations that include storage, a screen mount, tables, and screens.
  - Overview





  - Technical Specification

Parameter	Description
Screen 24” (x9)	23.8" 100Hz IPS Monitor
Screen 32” (x6)	31.5" 60Hz IPS Monitor
Mount & Storage	Screen mount (for screens), table x3, storage

2.3.3. Drone Set

- General Info

The **Drone Communication Set** extends the ROBOX system's communication range and situational awareness by adding an airborne relay and ISR capability.
It is based on the **EVO Max 4T** drone platform, equipped with a compact **BluSDR relay module** that links directly to the system's RF and IP-mesh network.
The drone is **launched from the container** and **controlled via its dedicated remote controller**, operating in either manual or semi-autonomous flight modes. Once airborne, it functions as a **communication bridge** between the deployed UGVs, the container, and remote operator stations—maintaining robust connectivity beyond line-of-sight and over complex terrain.
In addition to relay duties, the drone provides **real-time video and mapping data**, supporting reconnaissance, target confirmation, and mission oversight.
Compact, rapidly deployable, and fully integrated with the ROBOX communication architecture, the Drone COM Set enhances **range, persistence, and situational awareness** for all robotic assets operating in the field.

> The EVO Max 4T brings powerful technologies and advanced autonomy to users. EVO Max 4T assesses complex environments to create real-time 3D flight paths for unparalleled obstacle avoidance. Advanced sensors make flight in GNSS unavailable areas possible and the incredible new thermal payload unlocks new object identification and tracking scenarios. With a foldable, weather-resistant design, the EVO Max 4T is as portable as it is capable.

- Overview



Note: Drone with communication module assembly

- Technical Specification (Drone, for COM module see section 3.1)

The EVO Max 4T SkyLink 3.0 system comes with 6 antennas, 4 frequency bands, AES-256 encryption, and optional 4G integration to provide users with the most advanced flight capabilities ever in an EVO.



No Blind Spots



Superior Anti-Interference Capability



High-Precision Visual Navigation



A-Mesh Networking



8K 10x Optical Zoom



AI Target Recognition



Hot-Swappable Battery



15 km Transmission Range



42 Min Max. Flight Time



IP43 Weather Rating (Custom service)

15 km	1080P@30FPS	900MHz/2.4GHz /5.2GHz/5.8GHz*	< 150 ms
Image Transmission Distance	Real-Time Image Quality	Frequency Bands	Latency

2.3.4. ROCU-10 Controller

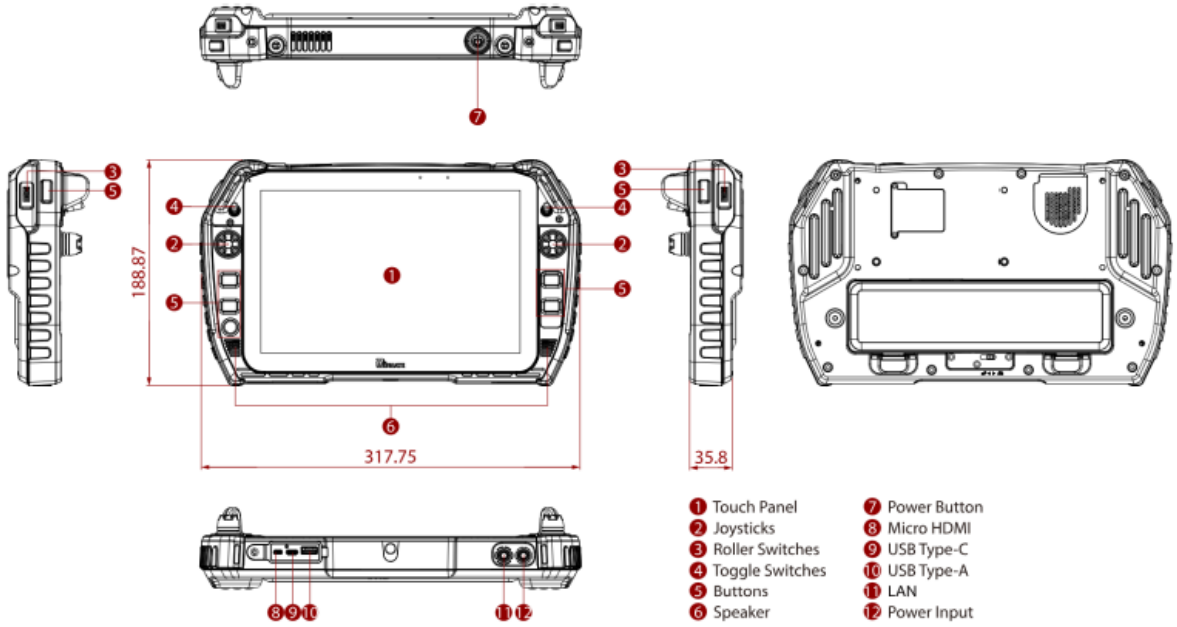
- General info

ROCU-10 controller allow the operator to drive the vehicle, operate the LEDs and scan the field using the four cameras.

- Overview



Dimensions: 317.75 x 188.87 x 35.8 mm



#	Parameter	Description
1	10.1" Touch Panel	Display and act as one of the inputs for the Controller.
2	Joysticks	An input device consisting of a stick that pivots on a base and reports its angle or direction to the device it is controlling.
3	Roller Switches	A round wheel connects the controller.
4	Toggle Switches	A type of electrical switch that is actuated by moving a lever back and forth to open or close an electrical circuit.
5	Buttons	Programmable function buttons

6	Speaker	Converting an electrical audio signal into a corresponding sound.
7	Power Button	Power on/off the controller.
8	Micro HDMI Output	Micro HDMI port connects to micro HDMI devices.
9	USB Output	USB 3.0 (Type-A)
10	USB Output	USB 3.0 (Type-C)
11	LAN	M10 LAN Connector (10/100 Mbps , Max to 250 Mbps)
12	DC power Input	100-240V, 50-60Hz / 19V DC

- Technical Specification

Parameter	Description
Display Specification	
Size	10.1"
Brightness	800 nits
Resolution	1920 x 1200
Contrast Ratio	800:1 (typ)
Viewing Angle	85/85/85/85
Touch	P-Cap Touch (10-point) with Anti-glare technology
Processor	Intel Core i5-1235U Processor (i5-1335U optional)
Operating System	Windows 10 IoT LTSC, Windows 11 IoT LTSC, Linux
System Memory	8GB DDR5 (up to 32GB)
Storage	128GB SSD (up to 2TB)
I/O Connectors	
Data Communication	1 x USB 3.0 Type A

Network Port	1 x LAN with M8 connector
Video Output	2 x Micro HDMI
Power Input	1 x USB Type C
SD Card	1 x Micro SD

Optional Wireless Communication

WLAN	802.11 a/b/g/n/ac/ax
Bluetooth	Bluetooth 5.2
GNSS	GPS / GLONASS
WWAN (optional)	LTE module
Custom RF	Optional

Others

Hotswap Battery	11.1 V, 6400mAh, 3S2P
Operating Temperature	-20~60C (AC); -10~50C (Battery)
Dimension (W x H x D)	319.94 x 188.87 x 51.5 mm (67.6 mm including Joystick)
Gross Weight	< 2 kg
Physical Channel	2x toggle switch; 2x joystick; 2x roller switch; 6x buttons; 2x speakers

Mechanical and Environment

IP Rating	IP65
Drop	MIL-STD-810H Method 516.8, Procedure IV, 4 feet
Shock	MIL-STD-810H Method 516.8, Procedure I
Vibration	MIL-STD-810H Method 514.8, Procedure I
EMC	CE, FCC, Compliance MIL-STD-461G

2.3.5. ROCU-1 Controller (optional)

- General Info

The ROCU-1 controller is used to drive the vehicle only and at close range (eye view).

- Overview



- Technical Specification

Parameter	Description
Dimensions (L x W x H)	60, 49, 184 mm (2.3", 1.9", 7.2")
Weight	0.8 kg (1.76 lbs.)
Communication	DIGI XBEE communication module
Power Supply	2 x AA 1.5v L91 lithium Batteries
Working Time	Max 8 Hours (operation mode dependent)
Operating Temperature	-20°C to +60° (-4°F to +140°F)
Environmental	Ingress Protection 65

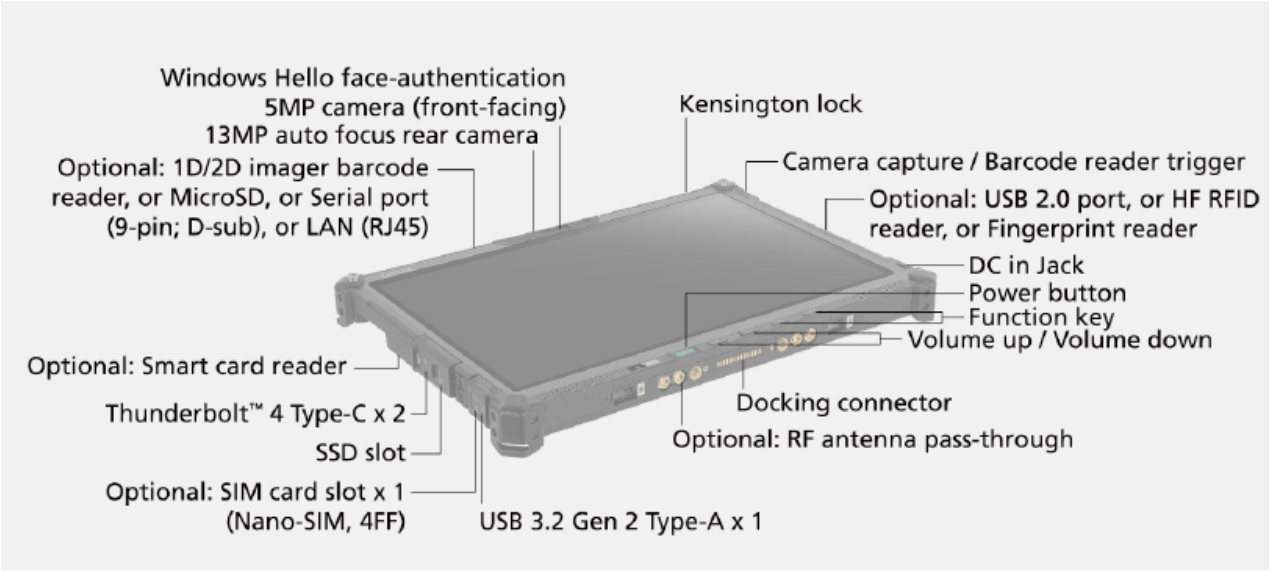
2.3.6. Commander Tablet

- General Info

Commander tablet used for commander observation and control on the mission when deployed outside the container (field-deployed).
The tablet connects to the communication operator box.



- Overview



- Technical Specification

Parameter	Description
Dimensions (W x D x H) & Weight	300 x 207 x 22.2mm (11.8" x 8.15" x 0.87") 1.41kg (3.11lbs)
Power	AC adapter (65W, 100-240VAC, 50/60Hz) Li-ion battery (7.74V, typical 4070mAh; min. 3800mAh) x 2 LifeSupport battery swappable technology Optional: High-capacity Li-ion battery (11.61V, typical 4070mAh; min. 3800mAh) x 2
Operating System	Windows 11 Pro
Display	12.2" Wide Viewing Angle TFT LCD WUXGA (1920 x 1200) Protection film

	1,200 nits LumiBond display with Getac sunlight readable technolo
Storage & Memory	16GB LPDDR5X (Memory on Package) Optional: 32GB LPDDR5X (Memory on Package) 256GB PCIe NVMe SSD Optional: 512GB / 1TB / 2TB PCIe NVMe SSD
	Integrated stereo speakers Integrated dual-array microphone
I/O Interface	Windows Hello face-authentication 5MP camera (front-facing) x 1 13MP auto focus rear camera x 1 DC in Jack x 1 USB 3.2 Gen 2 Type-A x 1 Thunderbolt™ 4 Type-C x 2 Docking connector x 1 Optional: SIM card slot x 1 (Nano-SIM, 4FF) iii Optional: Tri RF antenna pass-through for 4G (GPS, WLAN, WWAN) Optional: Quad RF antenna pass-through for 5G (GPS, WLAN, WWAN, WWAN MIMO)
Security Feature	TPM 2.0 Kensington lock Windows Hello face-authentication 5MP camera (front-facing) Getac Secure Erase Optional: Intel® vPro® Technology Optioanl: HF RFID reader, or Fingerprint reader viii, ix Optional: Smart card reader Optional: Absolute Persistence®
Rugged Feature	MIL-STD-810H certified IP66 certified MIL-STD-461G certified xi Vibration & 6ft (1.8m) drop resistant xii Optional: ANSI/UL 121201, CSA C22.2 NO. 213 (Class I, Division 2, Groups A, B, C, D) xiii Optional: Salt fog certified
Environmental Specification	Temperature xiv : - Operating: -29° C to 63° C / -20° F to 145° F - Storage: -51° C to 71° C / -60° F to 160° F Humidity: - 95% RH, non-condensing

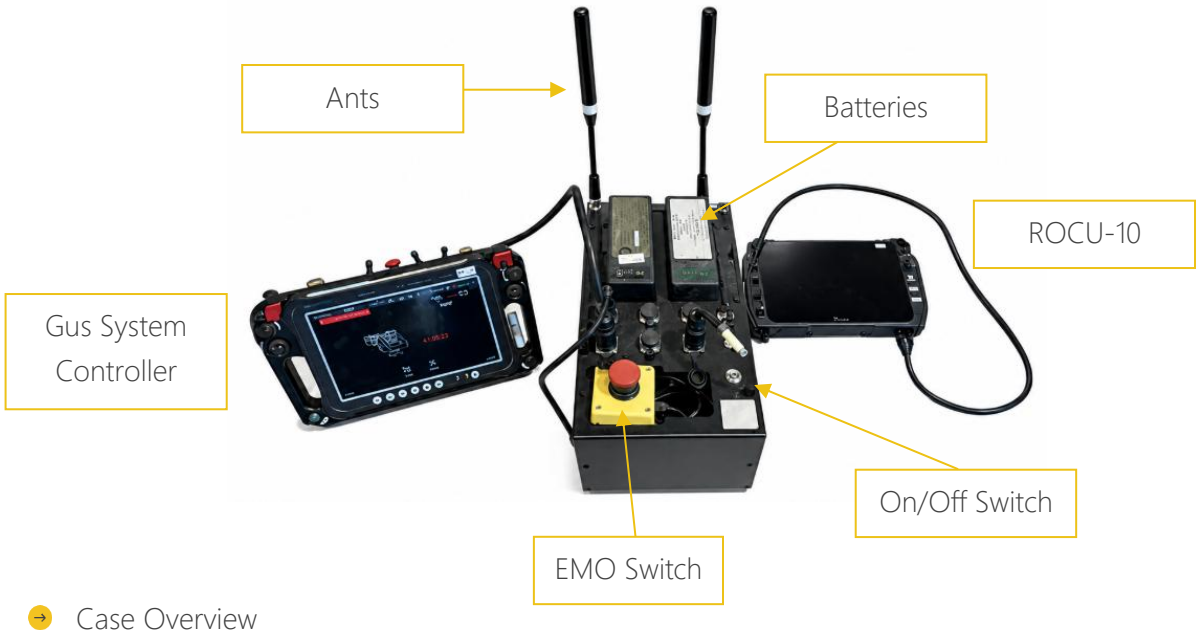
2.3.7. Operator Box

- General Info

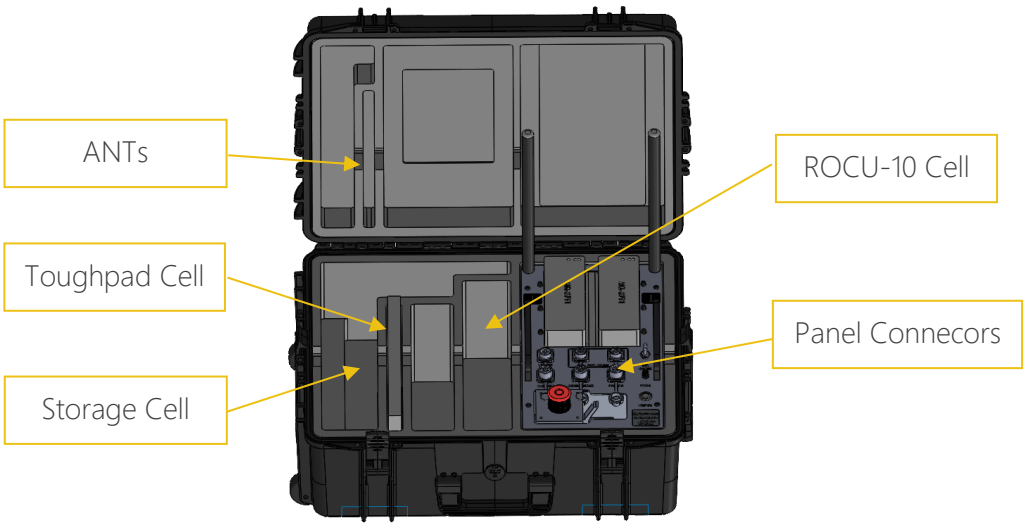
Operator box connected between the ROCU-10 controller and the gun system controller, to the controlled container/Tripod relay, and to system vehicles directly (For communication architecture, see section 3.6.4)

The box can be operated using two batteries or an external power connection.

- Box Overview



- Case Overview



- Technical Specification

Parameter	Description
Dimensions (L x W x H)	687x528x276 mm
Weight	10 kg
HW Interfaces internace	EMO ROCU-7, Pitbull controller ,2x barrey, External power, ETH /PWR, 2xETH, 2X RF ANT
Communication	DTC BLUESDR-30
Communication Range	1.5 km (to vehicle)
Power Supply	2X MR-2791 Batteries, External power
Working Time	Max 4 Hours (operation mode dependent)
Operating Temperature	-20°C to +60° (-4°F to +140°F)
Environmental	Ingress Protection 65

2.3.8. Tripod Relay Set

- General info

Relay for communication operator box for extended distance to container & vehicles.
Can be deployed anywhere.
Relay parts:

- Tripod for Communication box
- Relay Case for communication box
- Communication Box (RF, cellular, ANTs)

- Overview



Tripod



Relay case



Communication Box



- Technical Specification (Tripod)

Parameter	Description
Dimensions (L x W x H)	1.6X1.6X1.9 m
Max Height	3.5 m
Weight	5 kg
Max Load	50 kg
Mechanical Interface	Mount for 4X RF ANTs, Cellular & GPS ANT

- Technical Specification (Pelican components & Communication Box)

Parameter	Description
Dimensions (L x W x H)-Pelican	490x395x157
Dimensions (L x W x H)-COM Box	465x335x220
Weight (Pelican/COM box)	8kg/8kg
HW Interfaces interface	EMO,ROCU-10, Pitbull controller ,2x barrey, External power,3x ETH/PWR, 3xETH, 4X RF ANT, 4X Cellular ANT
Communication	IP MESH PHASE 5, Cellular
Communication Range (Phase 5)	5km
Power Supply	2X MR-2791 Batteries, External power
Working Time	Max 10 Hours (operation mode dependent)
Operating Temperature	-20°C to +60° (-4°F to +140°F)
Environmental	Ingress Protection 65

3. System Low Level Technical Specification

3.1. SW

3.1.1. General Info

The Robot Control and Fleet Management System is a distributed, cloud-agnostic software platform designed to enable real-time monitoring, control, and coordination of autonomous mobile robots. The system leverages ROS 2 (Robot Operating System 2) as its core middleware, providing a robust, scalable, and industry-standard framework for robotic applications.

- System Purpose: This software platform enables operators to:
  - Control individual robots remotely through an intuitive web-based interface
  - Monitor and manage multiple robots simultaneously as a coordinated fleet
  - Visualize real-time sensor data and telemetry from distributed robotic platforms
  - Maintain situational awareness across geographically distributed robot deployments

3.1.1.1. Key Capabilities

- Remote Robot Control:
  - Real-time manual control via virtual joystick interface
  - Autonomous navigation with waypoint planning
  - Emergency stop and safety override capabilities
- Fleet Management:
  - Health monitoring and predictive maintenance alerts
  - Battery management and charging coordination
  - Task assignment and workload distribution
- Sensor Integration:
  - Multi-camera video streaming with compression
  - IMU data fusion for accurate positioning
  - GPS integration for outdoor navigation
  - Extensible sensor framework for custom sensors
- Data Visualization:
  - Live camera feeds with overlay graphics
  - Performance metrics and analytics dashboards
  - Historical data playback and analysis
  - Customizable widgets and layouts
- Communication Infrastructure:
  - DDS-based publish-subscribe architecture for low-latency communication
  - Quality of Service (QoS) policies optimized for robotic applications
  - Automatic network discovery and robot registration

- Fault-tolerant communication with automatic reconnection
- Support for wired and wireless networks (WiFi, LTE, 4G,DTC)

- Target Application
  - This system is designed for a wide range of robotic applications:

3.1.1.2. System Benefits

- Hardware Agnostic Design:
  - Runs on diverse computing platforms (x86, ARM, Jetson)
  - Containerized architecture ensures consistency across deployments
  - No vendor lock-in for hardware components
  - Easy integration with existing robot platforms
- Scalability:
  - Horizontal scaling from single robot to large fleets
  - Modular architecture allows adding/removing components
  - Efficient resource utilization through containerization
  - Support for distributed computing across multiple machines
- Reliability:
  - Fault-tolerant design with automatic recovery
  - Redundant communication paths
  - Comprehensive error handling and logging
  - Proven ROS 2 middleware foundation
- Developer Friendly:
  - Standard ROS 2 interfaces for easy integration
  - Extensive API documentation
  - Python-based components for rapid development
  - Open architecture supporting custom extensions

3.1.1.3. System Characteristics

- Performance:
  - Control latency: < 100ms end-to-end
  - Sensor data rates: 10-100 Hz depending on sensor type
  - Support for 20+ robots per operator station
  - Real-time video streaming at 15-30 FPS
- Availability:
  - Designed for 24/7 operation
  - Automatic recovery from component failures
  - Graceful degradation under network stress
  - Health monitoring and alerting

- Maintainability:
  - Containerized deployment simplifies updates
  - Rolling updates minimize downtime
  - Comprehensive logging and diagnostics
  - Configuration management through YAML files
- Security:
  - DDS-Security for authenticated communication
  - Role-based access control for operators
  - Encrypted communication channels (optional)
  - Audit logging for compliance

3.1.1.4. System Constraints and Limitations

- Network Requirements:
  - Minimum bandwidth: 2 Mbps per robot for basic operation
  - Recommended: 5-10 Mbps per robot for full functionality
  - Maximum latency: 200ms for acceptable performance
  - Stable network connection required for optimal operation
- Hardware Requirements:
  - Platform Side: Jetson Xavier NX or higher (8GB+ RAM recommended)
  - Operator Side: Modern tablet/laptop (4GB+ RAM, dual-core processor minimum)
  - Storage: 20GB+ available for platform side, 5GB+ for operator side
- Operational Constraints:
  - Indoor/outdoor operation depends on sensor configuration
  - GPS required for absolute positioning in outdoor environments

3.1.2. SW Stack

3.1.2.1. Core Technologies

- ROS 2 (Robot Operating System 2):
  - Distribution: Humble Hawksbill (LTS) or Iron Irwini
  - Release: ROS 2 Humble (May 2022 - May 2027 support)
  - Language Bindings: Python 3.10+, C++ (optional for performance-critical nodes)
  - Middleware: DDS (Data Distribution Service)
  - Purpose: Core robotics middleware providing communication, device drivers, and libraries
- DDS Implementations:
  - Primary: CycloneDDS 0.10.x
    - ➢ High performance, lightweight

- Excellent real-time characteristics
- Low resource consumption

- Alternative: Fast-DDS 2.10.x

  - Feature-rich implementation
  - Good discovery mechanisms
  - Extensive QoS support

- Protocol: RTPS (Real-Time Publish Subscribe) over UDP/TCP

- Python Ecosystem:
  - Version: Python 3.10 or higher
  - Core Libraries:

    - rclpy - ROS 2 Python client library
    - numpy 1.23+ - Numerical computing
    - scipy 1.10+ - Scientific computing
    - opencv-python 4.7+ - Computer vision
    - pyserial 3.5+ - Serial communication
    - pyyaml 6.0+ - Configuration file parsing

3.1.2.2. Platform Side Software Stack

- Operating System:
  - Base OS: Ubuntu 22.04 LTS (Jammy Jellyfish)
  - Architecture: ARM64 (aarch64) for Jetson
  - Kernel: Linux 5.15+ with real-time patches (optional)
  - JetPack: 5.1+ for Jetson platforms (includes CUDA, cuDNN, TensorRT)

- Container Runtime:
  - Docker: 24.0.x or higher
  - Docker Compose: 2.20.x or higher
  - Base Images:

    - nvcr.io/nvidia/l4t-ros:humble-slam - NVIDIA L4T with ROS 2
    - GPU acceleration support built-in

- ROS 2 Packages:
  - Core ROS 2:

ros-humble-ros-base # Base ROS 2 installation ros-humble-rosidl-default-generators # Message generation ros-humble-rosidl-default-runtime # Message runtime ros-humble-rmw-cyclonedds-cpp # CycloneDDS middleware

- Navigation and Control:

ros-humble-nav2-bringup # Navigation 2 stack ros-humble-nav2-msgs # Navigation messages ros-humble-control-msgs # Control messages ros-humble-geometry-msgs # Geometry messages ros-humble-sensor-msgs # Sensor messages

- Sensors and Perception:

ros-humble-lidar-drivers # LiDAR device drivers ros-humble-image-transport # Efficient image transport ros-humble-cv-bridge # OpenCV-ROS bridge ros-humble-pcl-ros # Point Cloud Library integration

- Diagnostics and Tools:

ros-humble-diagnostic-msgs # Diagnostic messages ros-humble-diagnostic-updater # Diagnostic utilities ros-humble-rqt # ROS 2 GUI tools ros-humble-rviz2 # 3D visualization tool

- Hardware Interface Libraries:

  - CAN Bus: python-can 4.2+ - Controller Area Network support
  - GPIO: Jetson.GPIO - GPIO control for Jetson
  - Serial: pyserial 3.5+ - UART/Serial communication
  - Camera: v4l2 drivers, gstreamer 1.20+ for video streaming

3.1.2.3. Operator Side Software Stack

- Operating System:
  - Supported: Windows 10/11, macOS 12+, Ubuntu 20.04+
  - Architecture: x86_64, ARM64 (Apple Silicon supported)
  - Browser Requirements: Chrome 100+, Firefox 100+, Safari 15+
- Container Runtime:
  - Docker Desktop: 4.20+ (Windows/macOS)
  - Docker Engine: 24.0+ (Linux)
  - Docker Compose: 2.20+
- ROS 2 Packages:
  - Core Components:

ros-humble-ros-base # Base ROS 2 ros-humble-rosbridge-suite # WebSocket bridge ros-humble-rosapi # ROS API for rosbridge ros-humble-rmw-cyclonedds-cpp # DDS middleware

  - Message Types:

ros-humble-common-interfaces # Standard message types ros-humble-geometry-msgs # Geometry messages ros-humble-sensor-msgs # Sensor messages ros-humble-nav-msgs # Navigation messages

- Web Technologies:
  - Frontend Framework:

- Vue.js: 3.3.x - Progressive JavaScript framework
- Build Tool: Vite 4.4+ - Next generation frontend tooling
- Language: TypeScript 5.0+ or JavaScript ES2022+

UI Framework and Components:

- Tailwind CSS: 3.3+ - Utility-first CSS framework
- Vue Router: 4.2+ - Official router for Vue.js
- Pinia: 2.1+ - State management library
- Vueform: 1.2+ - Form builder and validation

2D Mapping:

- Leaflet: 1.9+ - Interactive maps
- OpenLayers: 7.4+ - High-performance mapping library

Charts and Graphs:

- Chart.js: 4.3+ - Simple yet flexible charting
- Apache ECharts: 5.4+ - Powerful charting library
- Plotly.js: 2.24+ - Scientific graphing

ROS Communication:

- roslib.js: 1.3+ - JavaScript library for ROS
- WebSocket API: Native browser support
- EventSource: Server-sent events for real-time updates

Web Server:

- Nginx: 1.24+ - High-performance HTTP server
- Static file serving for Vue.js application
- WebSocket proxy for rosbridge
- Load balancing (optional)
- SSL/TLS termination

Backend Services (Optional):

- Flask: 2.3+ - Python web framework for REST APIs
- FastAPI: 0.100+ - Modern Python web framework
- PostgreSQL: 15+ - Relational database for data logging
- Redis: 7.0+ - In-memory cache for session management

3.1.2.4. Development Tools

- Integrated Development Environments:
  - VS Code: 1.80+ with extensions:
    - ROS extension
    - Python extension
    - Docker extension
    - Vue - Official extension
- Build and Package Management:
    - colcon: 0.15+ - ROS 2 build tool
    - npm: 9.0+ - Node package manager
    - pip: 23.0+ - Python package installer
    - poetry: 1.5+ - Python dependency management (optional)
- Version Control:
  - Git: 2.40+
  - Git LFS: 3.3+ - Large file storage for datasets
  - GitHub/GitLab: Repository hosting and CI/CD
- Testing Frameworks:
  - Python Testing:
    - pytest: 7.4+ - Python testing framework
    - pytest-cov: 4.1+ - Coverage plugin
    - mock: 5.0+ - Mocking library
    - colcon test: ROS 2 testing integration
  - JavaScript Testing:
    - Vitest: 0.34+ - Vite-native test framework
    - Jest: 29.6+ (alternative) - JavaScript testing
    - Cypress: 12.17+ - End-to-end testing
    - Vue Test Utils: 2.4+ - Vue component testing
  - Continuous Integration/Deployment:
    - GitHub Actions - Automated workflows
    - GitLab CI/CD - Alternative CI/CD pipeline
    - Docker Hub - Container registry
    - Jenkins: 2.400+ (self-hosted option)

3.1.2.5. Testing Frameworks:

- Python Testing:
  - pytest: 7.4+ - Python testing framework
  - pytest-cov: 4.1+ - Coverage plugin
  - mock: 5.0+ - Mocking library
  - colcon test: ROS 2 testing integration
- JavaScript Testing:
  - Vitest: 0.34+ - Vite-native test framework
  - Jest: 29.6+ (alternative) - JavaScript testing
  - Cypress: 12.17+ - End-to-end testing
  - Vue Test Utils: 2.4+ - Vue component testing
- Continuous Integration/Deployment:
  - GitHub Actions - Automated workflows
  - GitLab CI/CD - Alternative CI/CD pipeline
  - Docker Hub - Container registry
  - Jenkins: 2.400+ (self-hosted option)
- 3.5 Monitoring and Debugging Tools
  - htop - Interactive process viewer
  - iotop - I/O monitoring
  - nethogs - Network bandwidth monitoring
  - docker stats - Container resource usage
- Network Analysis:
  - Wireshark 4.0+ - Network protocol analyzer
  - tcpdump - Packet capture tool
  - iftop - Network bandwidth monitoring
- Log Management:
  - journalctl - systemd journal viewer
  - Docker logs - Container log streaming
  - logrotate - Log rotation utility

3.1.2.6. Security and Compliance Tools

- DDS Security:
  - CycloneDDS Security - Authentication and encryption
  - OpenSSL: 3.0+ - Cryptographic library
  - Certificate Management - X.509 certificate tools
- Container Security:
  - Docker Scan - Vulnerability scanning

- Trivy - Container security scanner
- Clair - Static analysis for vulnerabilities

- Network Security:
  - iptables/nftables - Firewall configuration
  - fail2ban - Intrusion prevention
  - OpenVPN/WireGuard - VPN solutions

3.1.2.7. Documentation Tools

- Technical Documentation:
  - Sphinx: 7.0+ - Documentation generator
  - Doxygen: 1.9+ - Code documentation
  - Markdown - Lightweight markup language
  - PlantUML - Diagram generation
- API Documentation:
  - Swagger/OpenAPI - REST API documentation
  - ROS 2 Autodoc - Automatic ROS documentation
  - TypeDoc - TypeScript documentation generator

3.1.2.8. Software Stack Compatibility Matrix

Component	Platform Side	Operator Side	Notes
OS	Ubuntu 22.04 ARM64	Windows/macOS/Linux	Hardware agnostic
ROS 2	Humble/Iron	Humble/Iron	Must match versions
Python	3.10+	3.10+	3.10 minimum
Docker	24.0+	24.0+	Docker Desktop for Win/Mac
DDS	CycloneDDS/FastDDS	CycloneDDS/FastDDS	Must match implementation
Node.js	Not required	18+	For UI development
Browser	Not required	Chrome 100+/Firefox 100+	Modern browser required

- Open Source Licenses:
  - ROS 2: Apache 2.0
  - CycloneDDS: Eclipse Public License 2.0
  - Vue.js: MIT License
  - Python: PSF License
  - Docker: Apache 2.0
  - Nginx: BSD-2-Clause
- Compliance Notes:
  - All core dependencies use permissive licenses
  - No GPL dependencies in distributed binaries
  - Attribution requirements documented in NOTICE file
  - Commercial use permitted under current license stack

3.1.2.9. Version Management Strategy

- Release Channels:
  - Stable: Production-ready releases (quarterly)
  - Beta: Feature preview releases (monthly)
  - Development: Nightly builds (automated)
- Versioning Scheme:
  - Semantic Versioning (SemVer): MAJOR.MINOR.PATCH
  - Example: 2.1.3
    - MAJOR: Breaking changes
    - MINOR: New features (backward compatible)
    - PATCH: Bug fixes
- Supported Versions:
  - Current stable + 1 previous major version
  - Security patches for 2 years
- Control command flow

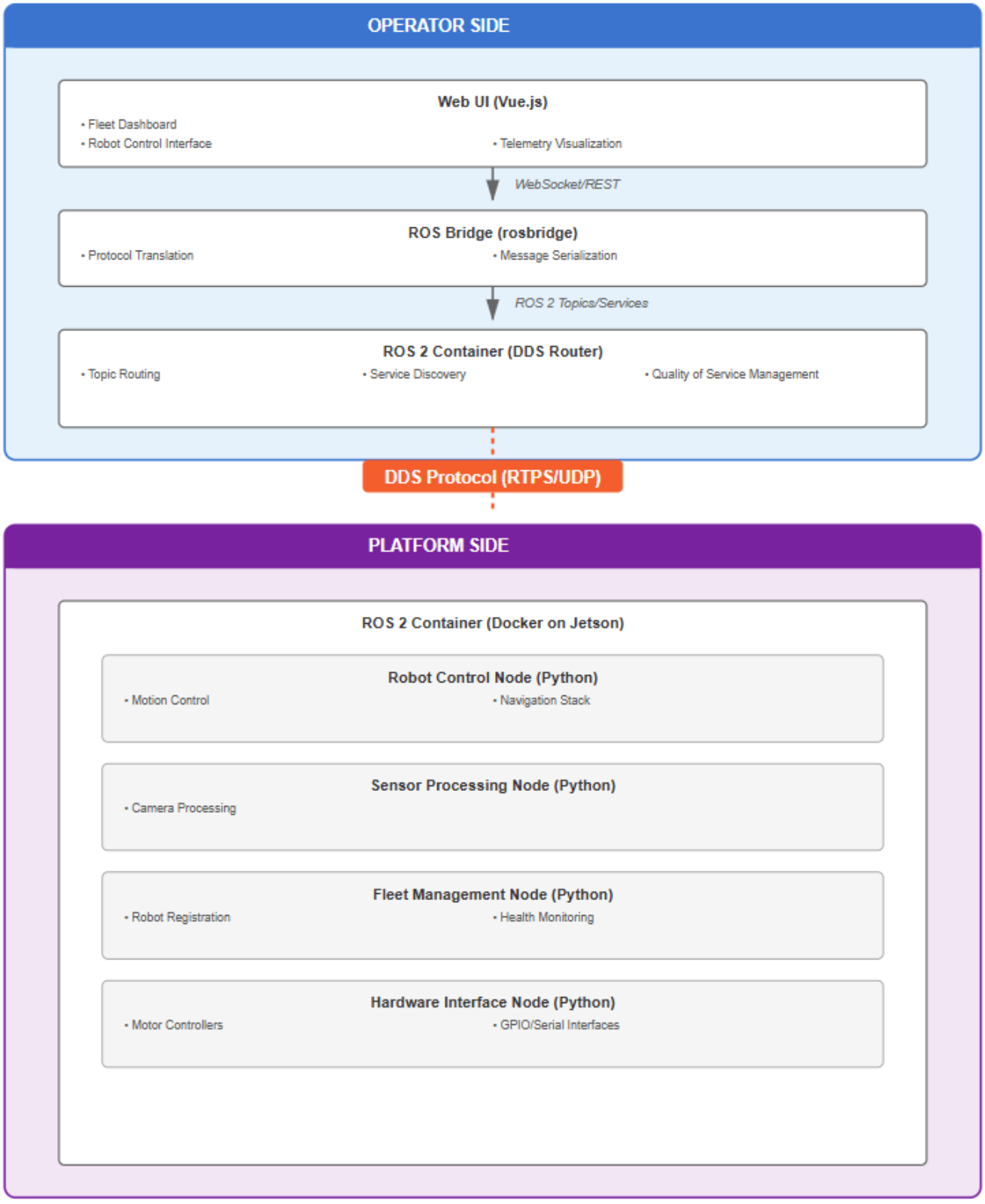
User Input (Joystick/Touch) ↓ Vue.js UI (Event Handler) ↓ WebSocket Message ↓ ROSBridge (Protocol Translation) ↓ ROS 2 Topic: /cmd_vel ↓ DDS Router (Topic Routing) ↓ DDS Network (RTPS/UDP) ↓ Robot Platform (DDS Listener) ↓ Robot Control Node (Python) ↓ Motor Commands Topic ↓ Hardware Interface Node ↓ Motor Controllers (CAN/Serial) ↓ Robot Actuators

- Sensor data flow

Robot Sensors (Camera/IMU) ↓ Sensor Drivers (ROS 2 Nodes) ↓ Sensor Processing Node (Python) ↓ Processed Sensor Topics ↓ DDS Network (RTPS/UDP) ↓ DDS Router (Operator Side) ↓ ROSBridge (Serialization) ↓ WebSocket Stream ↓ Vue.js UI (Visualization Components) ↓ User Display

- Security Specifications
  - Certificate Management:
    - Use X.509 certificates for authentication
    - Implement certificate rotation every 90 days
    - Maintain Certificate Revocation Lists (CRL)
  - Network Security
    - TLS 1.3 for WebSocket connections (production)
    - VPN tunnel for robot-operator communication (recommended)
    - Firewall rules limiting access to required ports only
    - Network segmentation between robot fleet and control network
  - User Authentication:
    - JWT tokens for web UI access
    - Token expiration: 24 hours
    - Refresh token mechanism
    - Role-based access control (RBAC)
- Maintenance and updates
  - Update Strategy:
    - Rolling updates for non-critical components
    - Scheduled maintenance windows for major updates
    - Rollback capability for failed updates
- Backup and Recovery
  - Backup Items:
    - Configuration files
    - Custom parameters
    - Mission logs
    - User settings
- Recovery Procedure:
    - Stop affected services
    - Restore configuration from backup
    - Verify configuration integrity
    - Restart services
    - Validate system operation

3.1.3. SW Architecture



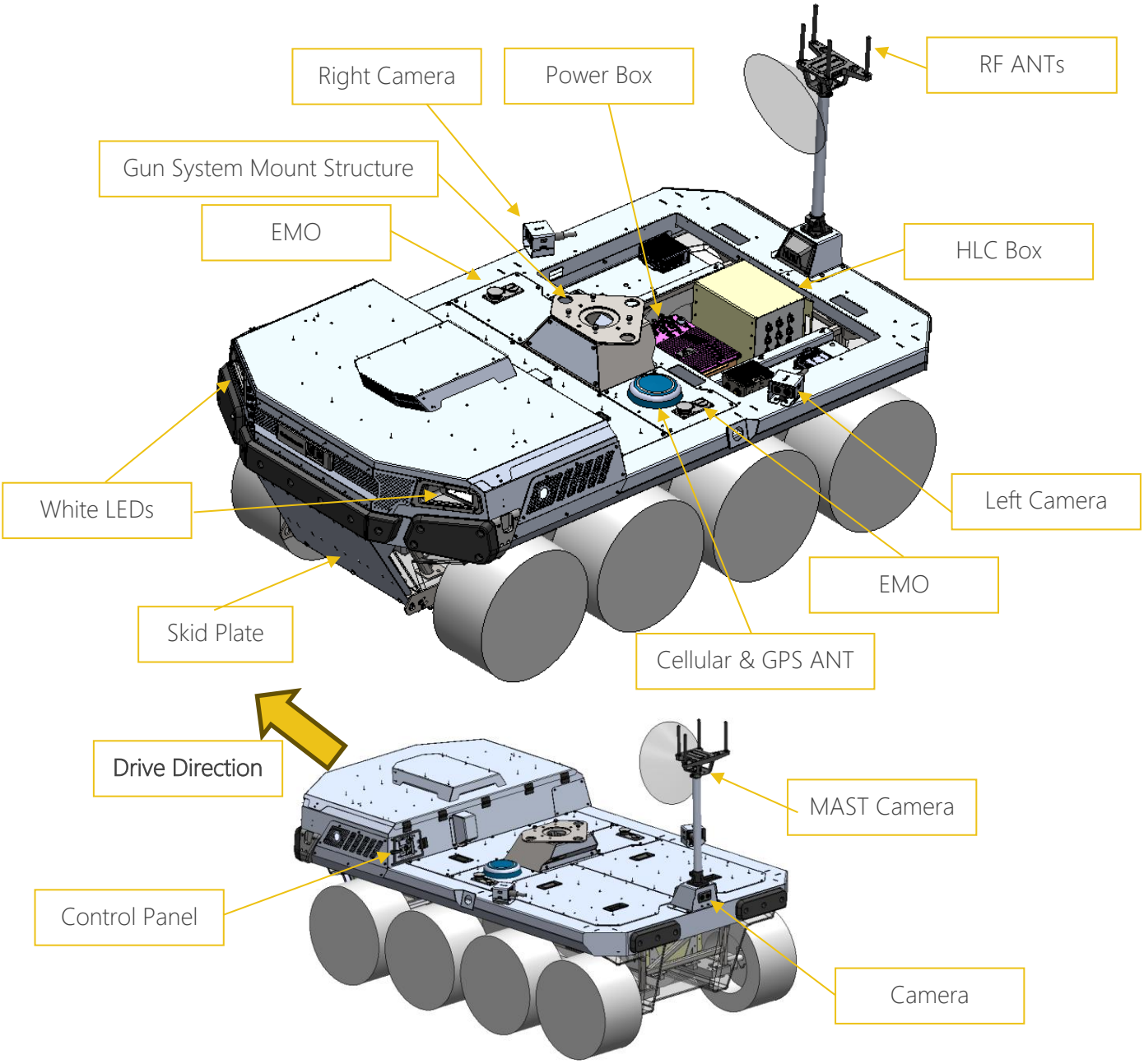
3.2. Vehicle Main Structure

3.2.1. General Info

We take the Argo vehicle and rework its internal body to fit our main structure system to make it a controlled and armed vehicle.

3.2.2. Overview

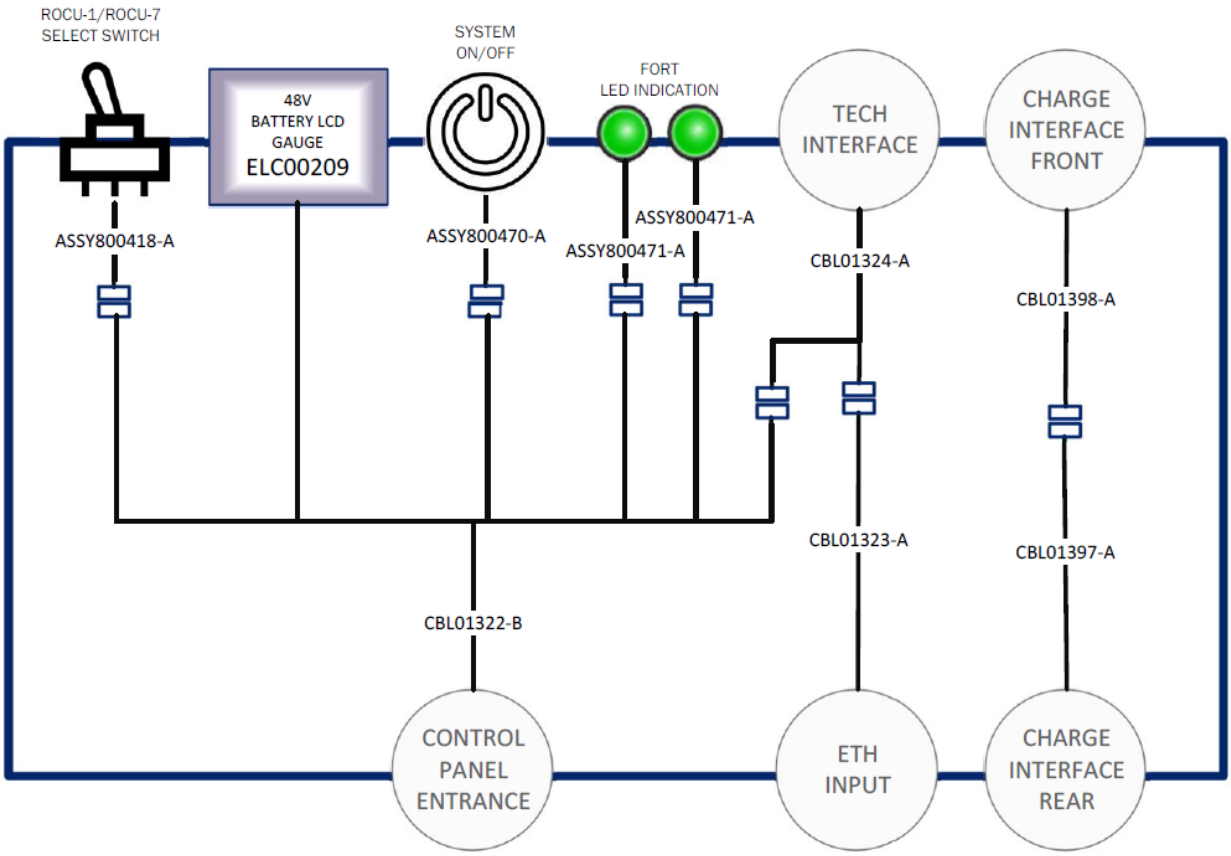
- Chassis and Components Overview
  - Main chassis assembled to the vehicle designed to support system components and gun system



- Front Panel Control

  The front panel control is located on the vehicles chassis and has a few features:

  - On/Of switch: to turn on and off the vehicle power (Low &High level control system, not engine)
  - Charge Connector: Interface to charge the internal battery
  - Toggle switch: to allow selection between ROCU-1 or ROCU-7 controllers
  - Tech Connector: interface to allow ETH and CanBus to the vehicle
  - SOC LCD screen: to show the battery status





3.3. EMO

3.3.1. General info

- System has two types of EMO: EMO Switch, wireless (Fort)
  - EMO Switch

Emergency Stop Pushbutton for all types of applications.
Lock lever design provides quick and easy installation.
Available vibration-resistant terminals (A22NE-P & A22NE-PD).
Lock detection function provides an added level of safety by minimizing emergency stop button failures from improper installation and damage.

  - Wirless (Fort,EPC-1000)

Parameter	Description
Certified Wireless Safety Command Sender & Receiver	The Endpoint Controller can send and receive up to two different SIL 3-certified safety commands (such as e-stop, crawl, pause, etc.) over Wi-Fi or Ethernet– ideal for supervision or teleoperation from an offsite location. As a sender: Send commands to up to 30 Endpoint Controller or NSC Pro receivers simultaneously. As a receiver: Receive commands from Endpoint Controller or Safe Remote Control Pro devices.
Remote Control Receiver	The Endpoint Controller acts as the receiver for FORT's Safe Remote Control Pro, via ISM radio or Bluetooth communication.
Integrate with Safety Inputs	Integrate with third-party input devices such as buttons, light curtains, sensors, fire alarms, etc. to trigger safety commands.
Flexible Integration	Mount on industrial equipment and use in a range of weather conditions with IP65 rating.
Tamper resistant hardware	Manage, configure, and pair your devices with FORT Manager cloud or desktop apps.
Built-in security	Secure protocol prevents unauthorized access during boot, updates, and configuration.

3.3.2. Overview

- EMO Switch



- Wirless EMO (Fort, EPC-1000)



3.3.3. Technical Specification

- EMO Switch

Parameter	Description
Dimensions (LxWxH)	72X45X45 mm
Type	Push Turn
Working Temperature	-20 to +70°C
Current Rating	5 A
IP Rating	IP69K

- Wireless EMO (Fort)

Parameter	Description
Device Form Factor	Machine-mountable transmitter/receiver
Dimensions	228mm x 174mm x 61mm | 8.99" x 6.85" x 2.4"
Enclosure Rating	IP65
Base Unit Weight	878g (1.9 lbs)
Operating Temps	-40°C to 85°C
Operating Voltage	8V to 32V DC
Wired Interface	Power, CAN, and Field Input/Output
CAN Supported Protocols	CAN J1939, CANopen
Connectors	2x M12 Ethernet connectors; 23-pin main integration connector; mini USB-B (service port)
Antenna Connectors	1 RP-TNC | 2 RP-TNC | 2 RP-TNC
Inputs	3 dual channels
Supply Input Voltage	12V or 24V nominal, 8 to 32V full range
Power Draw	2.18 - 2.78 W
Safety Input Voltage	Normal 8-32V, Safety State open circuit/Hi-Z
Safety Outputs	3 dual channels (1 dedicated for FORT Safe Remote Control Pro)
Safety Output Voltage	Matches Supply Input Voltage
Current Limit	750 mA max per output channel
Safety Output Type	3 dual-channel current sourcing
Safety Command Response Time	100ms avg tested plus network latency
Safety Timeout Period	Selectable: 250ms, 500ms, 750ms, 1000ms
Pairing	1 EPC : 30 EPC, 1 Safe Remote Control Pro : 30 EPC

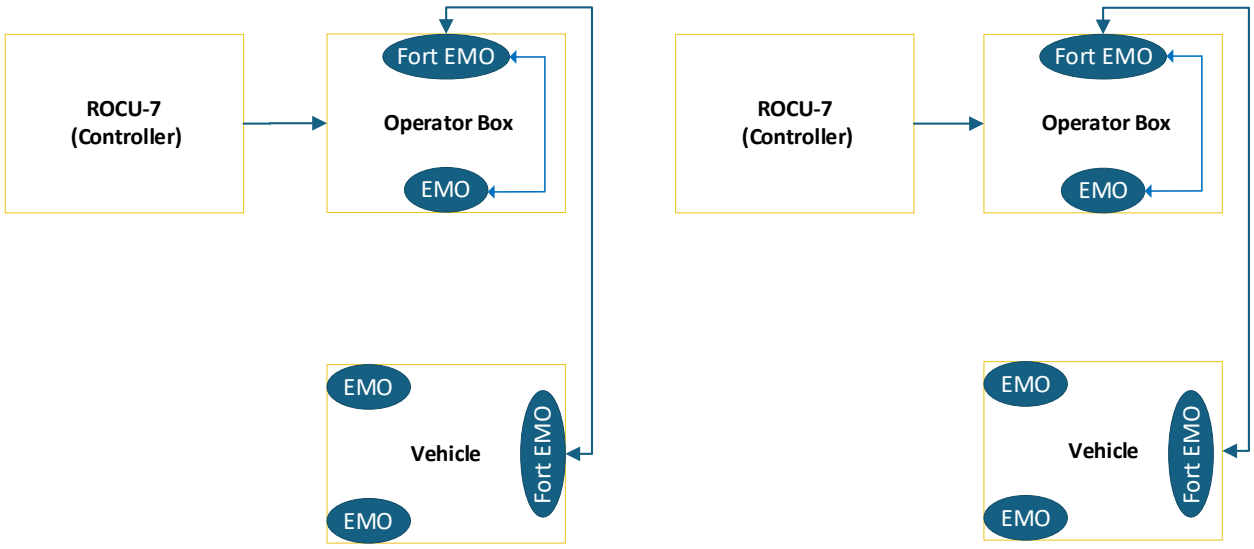
Security	Cryptographic boot, secure element, secure enclave, verification
Regulatory	FCC, IC compliant (US/CA), CE mark (EU)
Safety Certification	E-stop functionality Exida SIL 3 rating per IEC 61508

3.3.4. EMO Layout

- EMO Location

	EMO Switch	Wireless EMO (Fort)
Vehicle	2	1
Operator Box	1	1

- Layout



3.3.5. Operation

- EMO Switch

When pushing the EMO switch, the system brake (NC) comes into action, the vehicle gear goes to "N", and the motor turns off.

- Wireless EMO (Fort)

When pushing the EMO switch from the communication operator box, the wireless EMO goes into action, the vehicle system brake (NC) comes into action, the vehicle gear goes to "N", and the motor turns off.

3.4. Vehicle Brake

3.4.1. General Info and Operation

The vehicle has three parallel brake systems: an NC brake system, a hand brake, and a standard wheel brake system

- NC Brake System:

Overview

This mechanism is designed to ensure the remote-controlled vehicle comes to a complete stop in the event of a power loss, system malfunction, or emergency. It follows a Normally Closed (NC) safety methodology, meaning the brake is engaged by default when the system is unpowered or inactive.

Mechanical Design

The system consists of:

- Brake Lever: Mechanically connected to the vehicle's brake pedal.
- Return Spring: A high-tension spring connected to the brake lever, designed to pull the brake pedal into the braking position when released.
- Linear Actuator: An electrically powered actuator that stretches the spring to disengage the brake when the system is operational.
- Electromagnet Lock: Positioned at the end of the spring's travel, holding the spring in a stretched (brake-released) position as long as the system is powered.

Operating Principle

- Default State (No Power / System Off):

The electromagnet is de-energized, allowing the spring to contract. The spring pulls the brake lever, which presses the brake pedal, bringing the vehicle to a stop.

- Operational State (System On):

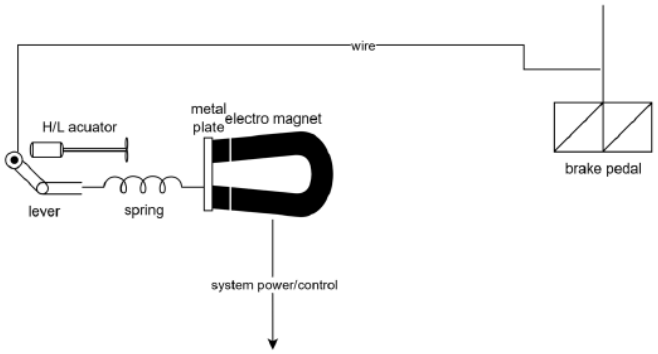
The linear actuator extends, stretching the spring. The electromagnet engages, holding the spring in the stretched position. This releases the brake and allows the vehicle to move.

- Fail-Safe Response:

In the event of a power failure, system malfunction, or intentional shutdown, the electromagnet releases the spring instantly. The stored spring tension engages the brake, stopping the vehicle immediately.

Safety Advantages

- Passive Safety: Brakes engage automatically without requiring power or control signals.
- Instant Response: Mechanical release of the spring ensures minimal delay in brake engagement.

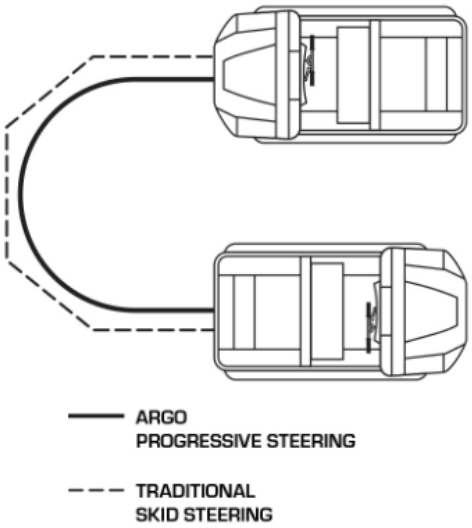


- Hand Brake System: To start movement, you must release the hand brake (Motor ignition will be allowed only when the hand brake is on).
- Standard Brake System: each axies (left, right) has its own brake and the vehicle control them both with one command (when you want to stop, you press on the brake), when turning to one side, the vehicle control each axis individualy to allow the vehicle turn (left side turn=right axis brake get into action\ right side turn=left axis brake get into action).

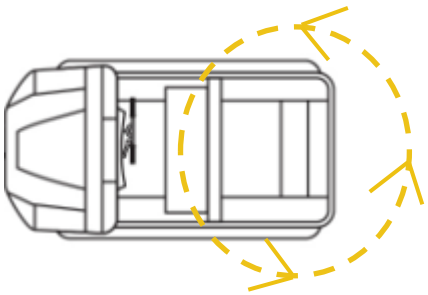
Steering overview:

ARGO PROGRESSIVE STEERING (APS)

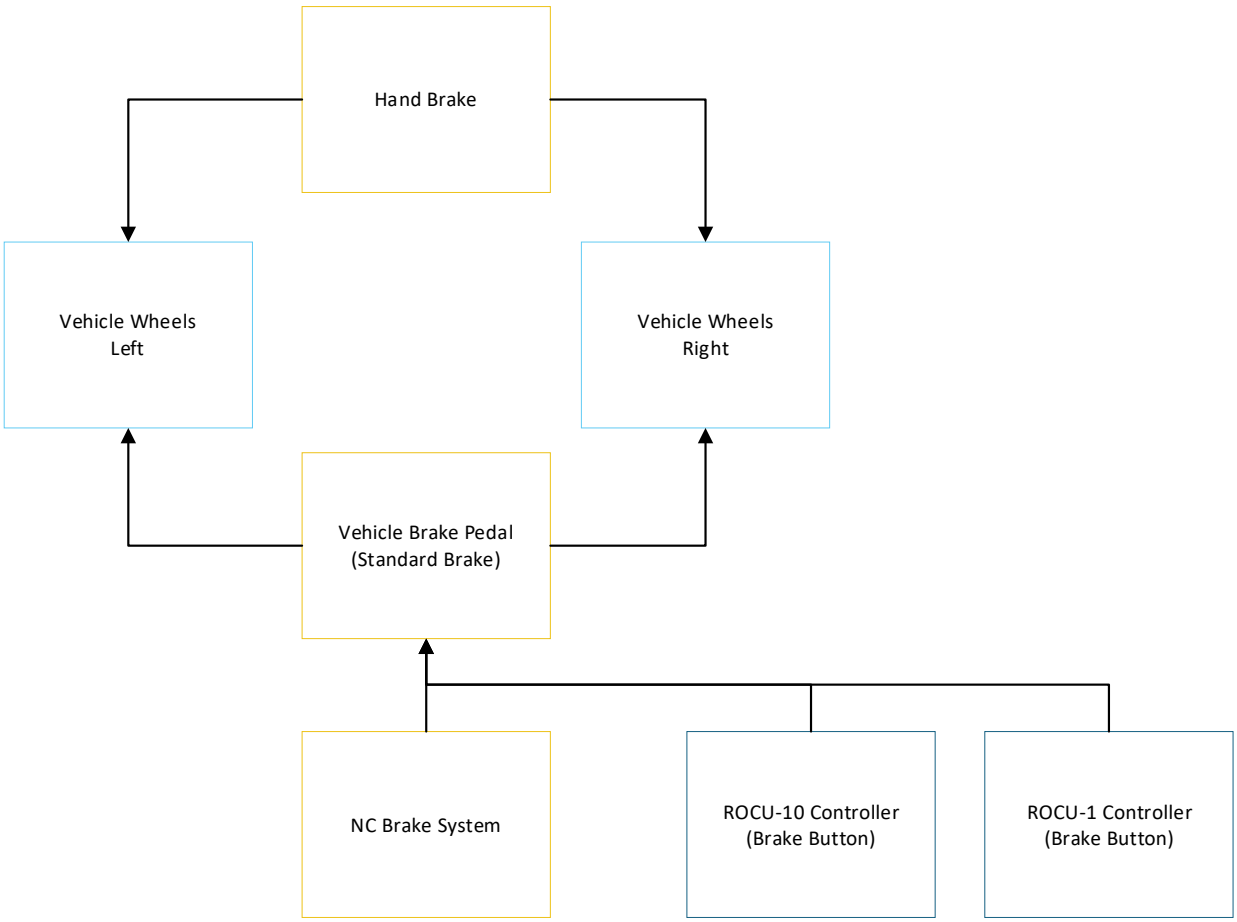
APS is an innovation in the area of braking force application to the transmission steering differential. By adding a spring element between the handlebars and the brake system we achieved a progressive, dual-rate steering system. Steering events are now effortless with an increase of 4X wider range of steering motion, offering the smoothest turning available in an ARGO XTV. APS makes it as fun and comfortable to drive as any UTV.



Note: ARGO vehicle can do pivot (turn on spot) in LOW gear status



3.4.2. Brake Layout



3.5. Energy

3.5.1. Container

3.5.1.1. Generator

- Overview



- Technical Specification

Parameter	Description
Dimensions (LxWxH)	512x700x721 mm
Oil Tank	1.1 Liter
Gas Tank	19.2 Liter
Phase	1
Amp	23.9
Voltage	230v
Frq	50 Hz
Power	5500 w
RPM	3600
Operation Time	6.5 hours
Operation Temperature	-25°C to 50°C
Outputs	3 Outlets 2x230V/16A and 1x230V/32A all outlets are share the generator total output capacity (5500W Max)
Operation Time	10 Hours (Full tank, Full functionality)

3.5.2. Vehicle

The UGV system comprises two primary power domains: a 12 V domain, based on a 60 Ah AGM battery and the external alternator, and a 48V domain, based on a LiPo battery with a nominal energy capacity of 2,400 Wh. 12V-to-48V charger provides the interface between the two domains, while the mission loads are powered from the 48V battery through 24 V and 28 V DC-DC converters.

3.5.2.1. External Motor Alternator

We add an external alternator for system power management, separate from the vehicle power.

ALTERNATOR OUTPUT **MAX: 100 AMPS @6000RPM, 34 AMP @1600RPM**



Parameter	Description
Voltage [V]	12
Amp. [A]	80
Size A [mm]	38.00
Size B [mm]	13.00
Size C [mm]	67.50
G [qty.]	4
Pulley	AP
D.1 [mm]	9.50
D.2 [mm]	10.50
D.3 [mm]	10.50
L.1 [mm]	165.00
Plug	PL_2004

3.5.2.2. Low Level Control Battery

- Overview



- Technical Specification

Parameter	Description
Voltage	12V
Battery Capacity	60 Ah
Cold Cranking Amps	680
Length	242 mm
Width	175
Height	190
Base Hold-down	B13
Terminal Types	1
Weight filled	17.5 kg

3.5.2.3. High Level Control Battery

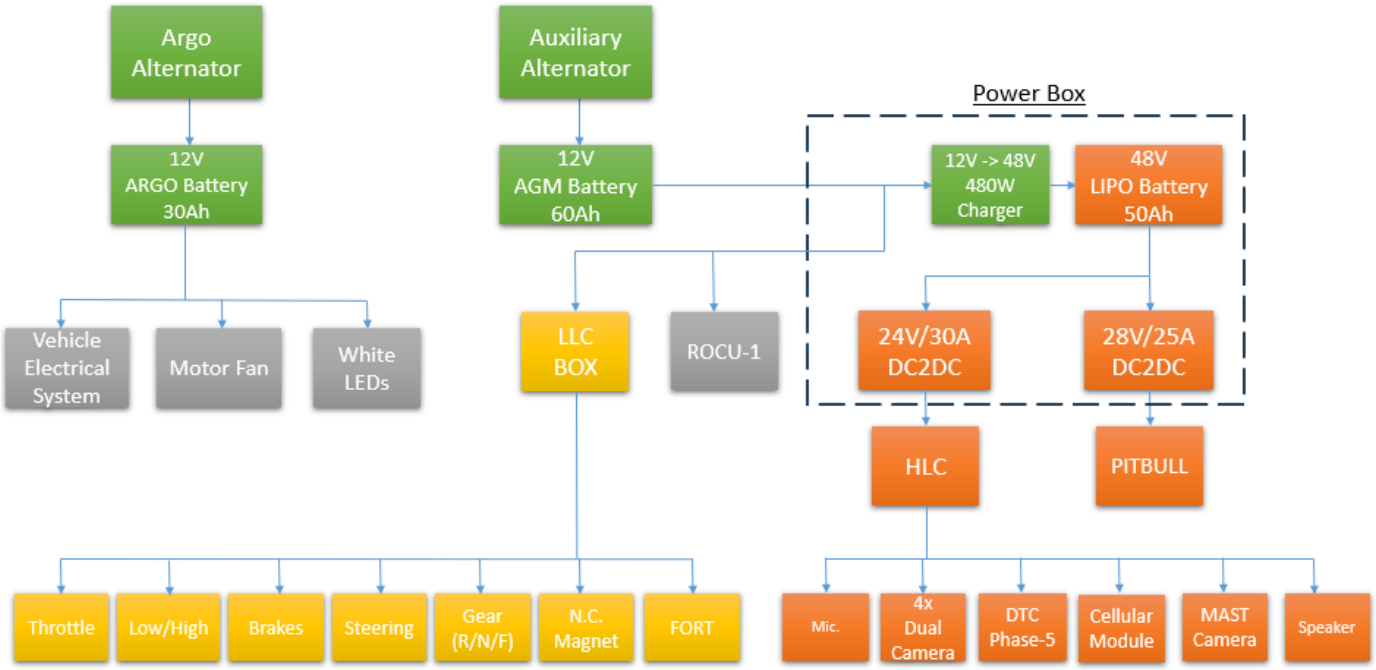
- Overview



- Technical Specification

Parameter	Description
Dimensions (LxWxH)	383X383X215 mm
Nominal Voltage	48 V
Weight	23 kg
Operation Temperature	-20°C to 60°C
Max Charge Current	50 A
Max Continuous Discharge Current	100 A
Cut off Voltage	37.5 V
Peak Current	150A for 1sec

3.5.2.4. Power Layout (Under test)



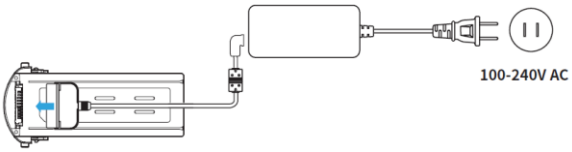
3.5.3. Drone Set

- General Info
  - Drone

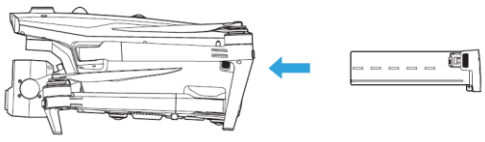
  Built-in battery
  - COM Module

  Powered from Drone

- Overview
  - Drone



Insert the battery into the aircraft



- COM Module



- Technical Specification
  - Drone

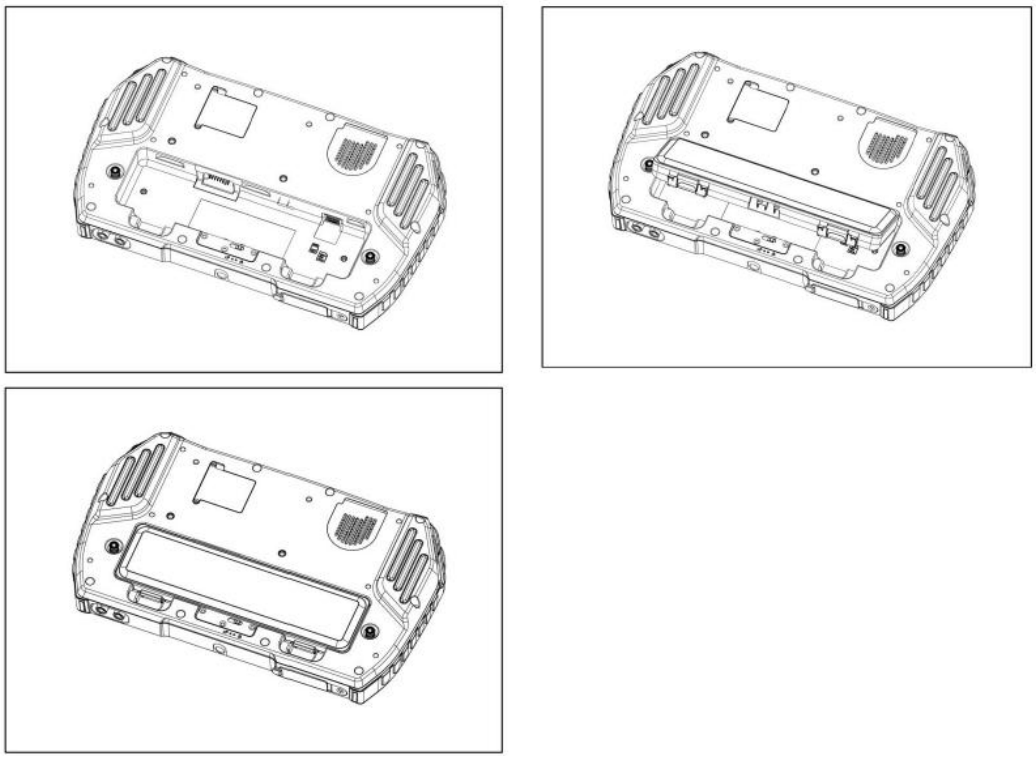
Parameter	Description
Battery Replacement	Built-in, not supported
Battery Type	Li-Po 3S
Rated Capacity	5800 mAh
Rated Energy	67 Wh
Voltage	11.55 V
Charging Time	About 120 minutes (Charging time depends on the actual remaining power)
Battery Endurance	Max brightness: 2.5 hours; 50% brightness: 4.0 hours

  - COM Module

Connection to Drone C-type input (5V→ DC-DC 12V)

3.5.4. ROCU-10 Controller (Optional)

Built-in battery



3.5.5. ROCU-1 Controller

2 x AA 1.5v L91 lithium Batteries



3.5.6. Operator Box

2x MR-2791 batteries or External power supply using AC-DC Adaptor

ELI-2791

Rechargeable Li-ion Battery Pack



General description

- Rugged, tough tactical battery pack
- High energy high Power density
- One wire battery communication technology
- Smart circuit providing remaining capacity, number of cycles and health status
- LED State Of Charge Indicator (SOCI)
- Low energy consumption during storage
- Easy integration into compact and light systems
- Meets civil and military standards
- Rugged sealed housing

Applications

Medical device, HLS, outdoor, energy backup, mobility aids (scooters), communications, military and more

Charge Method (CCCV)

- Charging through a desktop or tactical charger or through a dedicated charging board installed in host devise
- Charging Voltage: 33.6V / 16.8V
- Typical charging current: C/3

Safety

- Under/over voltage
- Under/over current
- Short circuit
- Temperature
- Reverse polarity

Quality System

- ISO-9001:2008; FDA & TUV Inspections

Standards

- UN-38.3
- MIL-STD-810E
- UL-1642 (for cells)

MSDS and Disposal

- Check with manufacturer

Electrical	
Capacity (typical)	15.6Ah / 20Ah @ 3A
Energy	230Wh / 290Wh
Max. voltage	16.8V
Nom. voltage	14.4V
Min. voltage	10V
Max. Dis. Current (cont.)	20A
Communication	One-wire; LED Fuel Gauge
Life Cycle	300 to 80% capacity @ 100% DOD

Mechanical	
Dimensions	(L;W;H) 186.2mm X 69.5mm X 65mm
Weight	1350 gr
Energy Density	223Wh/kg; 345Wh/l
Battery Housing	ABS
Color	Black, NATO Green, customable

Environmental	
Charge	0°C to 45°C
Discharge	-30°C to 60°C
Storage	-20°C to 50°C
Humidity	0 to 90%
Transportation	Class 9

MEAN WELL

160W AC-DC Reliable Green Industrial Adaptor

GST160A series

User's Manual

Video



UL62368-1 BS EN/EN62368-1 R33100 RoHS J62368-1 K60950-1 AS/NZS60950-1 GB4943.1 IS13252

MS IEC60950-1 (optional) IEC62368-1 TPTC004 Energy Verified Rendement Énergétique Vérifié

Features

- **Global certificates**
- Universal AC input / Full range
- 3 pole AC inlet IEC320-C14,Class I power unit
- Built-in active PFC function
- No load power consumption<0.15W
- **Energy efficiency Level VI**
- Comply with EISA 2007/DoE,NRCan,AU/NZ MEPS, EU ErP and CoC Version 5
- Protections: Short circuit / Overload / Over voltage / Over temperature
- -30~+70℃ wide range working temperature
- Fully enclosed plastic case
- LED indicator for power on
- Various DC plug quick adapter accessory available (Plug kit sold sperately, please refer to : https://www.meanwell.com/upload/pdf/DC_plug.pdf)
- 3 years warranty

Applications

- Consumer electronic devices
- Telecommunication devices
- Office facilities
- Industrial equipments

GTIN CODE

MW Search: https://www.meanwell.com/serviceGTIN.aspx

Description

GST160A is a highly reliable, 160W desktop style single-output green adaptor series. This product is a class I power unit (with FG), equipped with a standard IEC320-C14 AC inlet and adopting the input range from 85VAC to 264VAC. The entire series supplies different models with output voltages ranging between 12VDC and 48VDC that can satisfy the demands for various types of consumer electronic devices.
With the efficiency up to 94% and the extremely low no-load power consumption below 0.15W, GST160A is compliant with USA EISA 2007/DoE, Canada NRCan, Australia and New Zealand MEPS, Korea K-MEPS, EU ErP and Code of Conduct (CoC) Version 5. The supreme feature allows the adaptor to save the energy when it is either under the operating mode or the standby mode. The entire series utilizes the 94V-0 flame retardant plastic case. GST160A is certified for the international safety regulations.

3.5.7. Relay Pelican components

2x MR-2791 batteries or External power supply using AC-DC Adaptor

ELI-2791

Rechargeable Li-ion Battery Pack



General description

- Rugged, tough tactical battery pack
- High energy high Power density
- One wire battery communication technology
- Smart circuit providing remaining capacity, number of cycles and health status
- LED State Of Charge Indicator (SOCI)
- Low energy consumption during storage
- Easy integration into compact and light systems
- Meets civil and military standards
- Rugged sealed housing

Applications

Medical device, HLS, outdoor, energy backup, mobility aids (scooters), communications, military and more

Charge Method (CCCV)

- Charging through a desktop or tactical charger or through a dedicated charging board installed in host devise
- Charging Voltage: 33.6V / 16.8V
- Typical charging current: C/3

Safety

- Under/over voltage
- Under/over current
- Short circuit
- Temperature
- Reverse polarity

Quality System

- ISO-9001:2008; FDA & TUV Inspections

Standards

- UN-38.3
- MIL-STD-810E
- UL-1642 (for cells)

MSDS and Disposal

- Check with manufacturer

Electrical	
Capacity (typical)	15.6Ah / 20Ah @ 3A
Energy	230Wh / 290Wh
Max. voltage	16.8V
Nom. voltage	14.4V
Min. voltage	10V
Max. Dis. Current (cont.)	20A
Communication	One-wire; LED Fuel Gauge
Life Cycle	300 to 80% capacity @ 100% DOD

Mechanical	
Dimensions	(L;W;H) 186.2mm X 69.5mm X 65mm
Weight	1350 gr
Energy Density	223Wh/kg; 345Wh/l
Battery Housing	ABS
Color	Black, NATO Green, customable

Environmental	
Charge	0°C to 45°C
Discharge	-30°C to 60°C
Storage	-20°C to 50°C
Humidity	0 to 90%
Transportation	Class 9

MEAN WELL

160W AC-DC Reliable Green Industrial Adaptor GST160A series



User's Manual

Video

UL62368-1 BS EN/EN62368-1 R33100 RoHS J62368-1 K60950-1 AS/NZS60950-1 GB4943.1 IS13252

MS IEC60950-1 (optional) IEC62368-1 TPTC004 Energy Verified Rendement Énergétique Vérifié

Features

- **Global certificates**
- Universal AC input / Full range
- 3 pole AC inlet IEC320-C14,Class I power unit
- Built-in active PFC function
- No load power consumption<0.15W
- **Energy efficiency Level VI**
- Comply with EISA 2007/DoE,NRCan,AU/NZ MEPS, EU ErP and CoC Version 5
- Protections: Short circuit / Overload / Over voltage / Over temperature
- -30~+70℃ wide range working temperature
- Fully enclosed plastic case
- LED indicator for power on
- Various DC plug quick adapter accessory available (Plug kit sold sperately, please refer to : https://www.meanwell.com/upload/pdf/DC_plug.pdf)
- 3 years warranty

Applications

- Consumer electronic devices
- Telecommunication devices
- Office facilities
- Industrial equipments

GTIN CODE

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Description

GST160A is a highly reliable, 160W desktop style single-output green adaptor series. This product is a class I power unit (with FG), equipped with a standard IEC320-C14 AC inlet and adopting the input range from 85VAC to 264VAC. The entire series supplies different models with output voltages ranging between 12VDC and 48VDC that can satisfy the demands for various types of consumer electronic devices.
With the efficiency up to 94% and the extremely low no-load power consumption below 0.15W, GST160A is compliant with USA EISA 2007/DoE, Canada NRCan, Australia and New Zealand MEPS, Korea K-MEPS, EU ErP and Code of Conduct (CoC) Version 5. The supreme feature allows the adaptor to save the energy when it is either under the operating mode or the standby mode. The entire series utilizes the 94V-0 flame retardant plastic case. GST160A is certified for the international safety regulations.

3.6. Communication

3.6.1. RF Communication

3.6.1.1. General Info

- Communication modules by B.A. MICROWAVES (DTC Manufacture)
- We use two types:
  - BLUESDR-30 RADIO CLAMSHELL 1.98-2.70GHZ, 2W:
    for operator box
  - IP MESH PHASE 5 R.M,2X5W,BAMZ,1.98-2.70GHZ:
    for Tripod Relay and BISON vehicle

3.6.1.2. Overview

- DTC BLUESDR-30



- IP MESH PHASE 5



3.6.1.3. Technical specifications

- DTC BLUESDR-30

Parameter	Description
IO (Clamshell Core)	
RF COFDM transceiver 1	MMCX 50Ω (f)
RF COFDM transceiver 2	MMCX 50Ω (f)
GPIO and power out	6-way 1mm JST
Gigabit Ethernet	8-way 1mm JST
USB1 control and download	5-way 1mm JST
USB2 control and download	USB-C (PD)
Power input	4-way 2mm JST / USB-C with power delivery (USB-C PD)
Analogue audio	6-way 1mm JST
RS-232	3-way 1mm JST
IO (Plain Boxed)	
RF COFDM transceiver A	SMA 50Ω (f)
RF COFDM transceiver B	SMA 50Ω (f)
Power input	4-way Lemo (f)
USB2 with power delivery	USB-C PD
Gigabit Ethernet	RJ45
USB1 control and download	USB Type-A
USB2 control and download	USB Type-C
Analogue audio	3.5mm jack
GPIO and RS-232 serial	6-way Lemo (f)

Audio	
Talkback audio	Analogue mic/head with bias
Compression	G726 32kbit audio 8kHz sampling and mute
Input level (600Ω impedance)	2.5mV to 2.3V RMS
Output level	32mV RMS max into a 16Ω load
Data	
Data configuration	1k2 to 115k2, 7/8 bit, no/odd/even parity
Data interface	RS-232 or USB peripherals
Encryption	
Standard	DES
Licensed	AES128/256 (subject to export control)
Storage	
Medium	Internal microSD 128GB
Physical (Clamshell Core)	
Dimensions	95mm x 60mm x 15mm
Weight	110g
Dimensions	150mm (160mm incl. connectors) x 75mm x 24mm
Weight	462g approx.
Dimensions	157mm (165mm incl. mounting) x 75mm x 21mm
Weight	430g approx.
Dimensions	95mm x 60mm x 15mm
Weight	110g

IP MESH PHASE 5

Parameter	Description
Interfaces	
COFDM RF interfaces	N-Type x4 female (2 x Tx/Rx, 2 x Rx)
GPS antenna interface	SMA female
Power and Ethernet (Eth0)	6-way Amphenol 38999 series 3
Ethernet (Eth1)	RJ45
Config, data and audio	22-way Amphenol 38999 series 3
Composite video and power (optional)	4-way Amphenol 62GB series
SD/HD-SDI video	BNC female 75Ω
USB	Type A
IP Interface	
Primary and secondary Ethernet	100/1000Base-T
IP address allocation	DHCP dynamic IP addressing / static IP
RF Interfaces	
Antenna A	Channel 1 receive only
Antenna B	Channel 1 switched transmit/receive
Antenna C	Channel 2 receive only
Antenna D	Channel 2 switched transmit/receive
RF and Modulation	
Output frequency	Frequency variant dependent
Tuning step size	125kHz step
Output power	+37dBm per channel in 0.25dB step (10W total)

Bandwidth (MHz)	1.25, 1.5, 1.75, 2.5, 3.0, 3.5, 4.0, 5.0, 6.0, 7.0, 8.0, 10.0, 12.0, 14.0, 16.0, 20.0
Mesh capacity	Up to 87Mbps MIMO, 17Mbps standard Mesh
Modulation	COFDM 360 carrier modulation
Carrier modulation	BPSK / QPSK / 16QAM / 64QAM (adaptive)
FEC rate	FEC 1/2, FEC 2/3 (adaptive)
Receive diversity	Maximum ratio combining
Receive sensitivity	-98dBm (BW 2.5MHz QPSK 1/2)
Receive input level	+20dBm max.
Typical Range	
NLOS light urban	2200m †
LOS (e.g. ground to air)	170km †
Note	† Dependent on antenna height and gain
Streaming	
Format	UDP multicast/unicast; RTSP/RTP/UDP multicast/unicast
MJPEG	TCP/HTTP
Video	
Video input	Two video streams (with optional composite input); Max total throughput 1920x1080p30; Currently both video input resolutions must be identical
SDI input formats	1920x1080i 60/59.94/50Hz; 1920x1080p 30/29.97/25/23.97Hz; 1280x720p 60/59.94/50Hz; 720x576 50Hz; 720x480 59.94Hz
Composite input formats (optional)	PAL, NTSC
H.264 compression	AVC / H.264 / MPEG-4 Part 10; High profile level 4.0
Coding options	Horizontal scaling 3/4, 2/3, 1/2; Vertical scaling 1/2, 1/4; Sub-frame rate 1/2, 1/4, 1/8, 1/24
Encoder delay	1s to 10ms (mode dependent)

Encoder bitrates	0.25Mbps to 32Mbps
Digital audio input	SD/HD-SDI digital stereo pair
Sample rate	16kHz–48kHz
Coding modes	4 channel stereo or mono; MPEG audio Layer 1 64–448kbps; MPEG audio Layer 2 32–384kbps; MPEG audio Layer 3 8–256kbps
Talkback audio	
Compression	Microphone level/headphone output; G726 32kbit audio 8kHz sampling and mute
Input level (600Ω impedance)	0.25mV to 1.14V RMS
Output level	32mV RMS max into a 16Ω load
Store and Forward Options	
Storage format	SD card interface (Secure Digital) – not user accessible
Record options	Continuous or triggered
Files download	From web browser interface / RTSP
Video and audio clip size	30 seconds
Encryption	
Standard	DES
Licensed	AES128/256 (subject to export control)
GPS	
Built-in GPS receiver	Garmin GPS-15x-HW

3.6.2. Cellular & GPS Communiction

3.6.2.1. General Info

RUTX12

DUAL LTE CAT 6 INDUSTRIAL CELLULAR ROUTER

This powerful dual LTE Cat 6 router is a great fit for mission-critical applications. Two cellular modems for dual simultaneous connections, seamless mobile service switching, and load balancing features make this device irreplaceable in applications where losing connection is not an option.

3.6.2.2. Overview



3.6.2.3. Technical Specification

Parameter	Description
Manufacture	Teltonika
Mobile Module	2 x 4G LTE Cat 6 up to 300 DL/50 UL Mbps; 3G up to 42 DL/ 5.76 UL Mbps
Wireless Mode	802.11b/g/n/ac Wave 2 (Wi-Fi 5) with data transmission rates up to 867 Mbps (Dual Band, MU-MIMO), 802.11r fast transition, Access Point (AP), Station (STA)
Wi-Fi Security	WPA2-Enterprise: PEAP, WPA2-PSK, WPA-EAP, WPA-PSK, WPA3-SAE, WPA3-EAP, OWE; AES-CCMP, TKIP, Auto-cipher modes, client separation, EAP-TLS with PKCS#12 certificates, disable auto-reconnect, 802.11w Protected Management Frames (PMF)
SSID/ESSID	ESSID stealth mode
Wireless Connectivity Features	Wireless mesh (802.11s), fast roaming (802.11r), Relayd, BSS transition management

	(802.11v), radio resource measurement (802.11k)
Bluetooth	Bluetooth low energy (LE) for short-range communication
Internet-VAN	1 x WAN port 10/100/1000 Mbps, compliance with IEEE 802.3, IEEE 802.3u, 802.3az standards; supports auto MDI/MDIX crossover
Internet-LAN	4 x LAN ports, 10/100/1000 Mbps, compliance with IEEE 802.3, IEEE 802.3u, 802.3az standards, supports auto MDI/MDIX crossover
Input voltage range	9 - 50 VDC, reverse polarity protection, voltage surge/transient protection
Power consumption	Idle: 4 W, Max: 22 W

3.6.3. ANTs

3.6.3.1. RF

- Overview
  - ANT For Operation Box



  - ANT for Vehicle, Relay and MAST



- Technical Specification
  - ANT For Pelican Communication Support Box

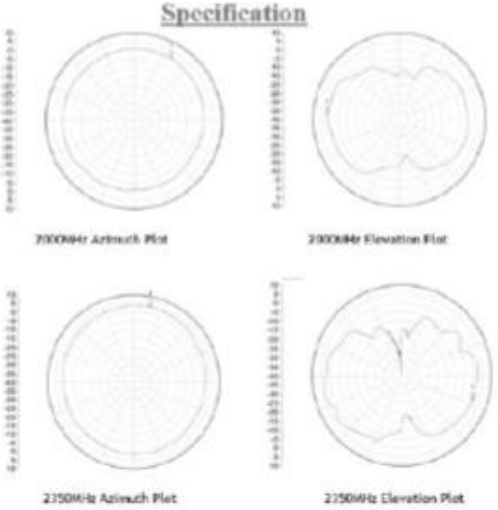
5dBi (Gooseneck) Omni- Directional Antenna

Gooseneck Omni directional antennas are ideal for multipath environments where RF penetration is critical. The Gooseneck also allow for placement on molly systems for bodyworn radio systems. Also applicable for larger type radios and UAV/USV type requirements.

The rugged antenna is made from machined aluminium, with a tubular glass fibre cover. Available in TNC, N Type or BNC connectors. Please note image may not be to size as frequency variations will occur. Other frequencies available including L, C, and LTE bands. Others on request

Frequency Range:	1980MHz - 2700MHz
Pattern	Co-linear omni-directional
Azimuth beamwidth	360degs
Elevation beamwidth	20degs +/-
Gain:	5dBi peak gain 1.98-2.20Ghz (1.3dBi-2.8dBi) 2.20-2.50GHz (2.8dBi- 5dBi) 2.50-2.70GHz (1.5dBi-3.0dBi)
VSWR:	2:1 maximum
Polarization:	Linear Vertical
Power	25w
Mechanical Form:	Tube with flex
Size:	303mm x 20mm plus connector
RF Connector:	TNC (m)
Colour:	Black
Weight	140g approx
Options:	N Type (m)

Specification





Unit/Part Number	Description
ANT-5GN-198-270	5dBi (GN) Omni Directional Antenna (1.98-2.70GHz)

- ANT for Vehicle, Relay and MAST

5dBi Flexible Antenna (1980-2700MHz)

apella solutions

sales@apella.co.uk
www.apellasolutions.com

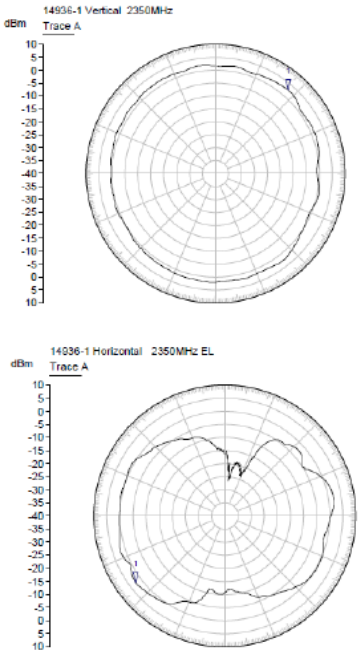
EXIBLE ANTENNA



Image for illustration only. Not to scale

Apella Solutions flexible antennas provides an omni-directional pattern with our latest addition to the Apella antenna family. Various frequencies are also available including L, S and C bands. Various connectors available from SMA, N(m)/N(f) type and TNC/BNC(m)

Frequency Range:	1980-2700MHz
Pattern:	Omni Directional
Azimuth Beamwidth:	360deg
Horz Beamwidth:	+30/-30 deg typical
Gain:	5dBi +/-2dB
VSWR:	2.5:1 minimum 2:1 nominal
RF Power Handling:	25W
Polarization:	Linear Vertical
Mechanical Form:	Flexible whip
Size:	125mm approx + connector
RF Connector:	N-Type(m) others available
Weight:	70 grams approx
Colour:	Black



3.6.3.2. Cellular & GPS (Bison Vehicle Set Side)

- Overview



Technical Specification

Parameter	Description
Cellular	
Antenna elements	4 elements
Cellular frequencies	4.4dBi @ 617–960MHz; 7.8dBi @ 1710–2700MHz; 7.9dBi @ 3400–4200MHz; 8.2dBi @ 5000–6000MHz
Cellular bands	LTE bands B1 to B86 (except B31, B72, B73, B87, B88); 5G bands n1 to n99
VSWR	< 2.5 over 95% of the band
Feed power handling	10W
Input impedance	50 Ω
Polarisation	Linear
Ground plane	Not required
Wi-Fi	
Antenna elements	2 elements
Peak Gain & Frequencies	5.0dBi: 2400–2500MHz; 7.5dBi: 5000–6000MHz
VSWR	< 2.5
Feed power handling	10W
Input impedance	50 Ω
Polarisation	Linear
Cable (GPS)	
Type	RG-174
Loss	3.4 dB/m @ 1000 MHz; 4.9 dB/m @ 1800 MHz
Diameter	0.1" / 2.7 mm
Jacket	Half matt PVC, UV resistant

Termination	QMA male, SMA male
Environmental, compliance	
IP rating	IP68
Operating temperature	-40° to 176°F / -40° to 80°C
Storage temperature	-40° to 176°F / -40° to 80°C
Compliance	RoHS, REACH, WHEE
Wind Survivability	137 mph, 220 km/h
Enclosure flammability	UL 94 HB
Cable flammability	UL 758 (VW-1)
UV resistance	UL 746C (F1 long-term UV exposure)
Salt Spray	MIL-STD 810F / ASTM B117
GPS	
Frequency range	1561–1602 MHz
Peak Gain & Frequencies	0.5 dBi @ 1575 MHz; 1.6 dBi @ 1602 MHz
VSWR	< 2 dB
Output return loss	10 dB max
LNA Gain	28 ±3 dB
Noise figure	1.5 dB max at 3.3V
Operating Voltage	3.3V
Power consumption	8.5 ±2.5 mA at 3.3V
Cable (LTE / 5G / Wi-Fi)	
Type	CFD-200
Loss	0.33 dB/m @ 900 MHz; 0.49 dB/m @ 2000 MHz; 0.55 dB/m @ 2500 MHz; 0.87 dB/m @ 5800 MHz

Diameter	0.2" / 5.0 mm
Jacket	Half matt PVC, UV resistant
Cellular Termination	QMA male, SMA male
Wi-Fi Termination	QMA male, RP-SMA male

3.6.3.3. Cellular & GPS (Relay Set Side)

- Overview

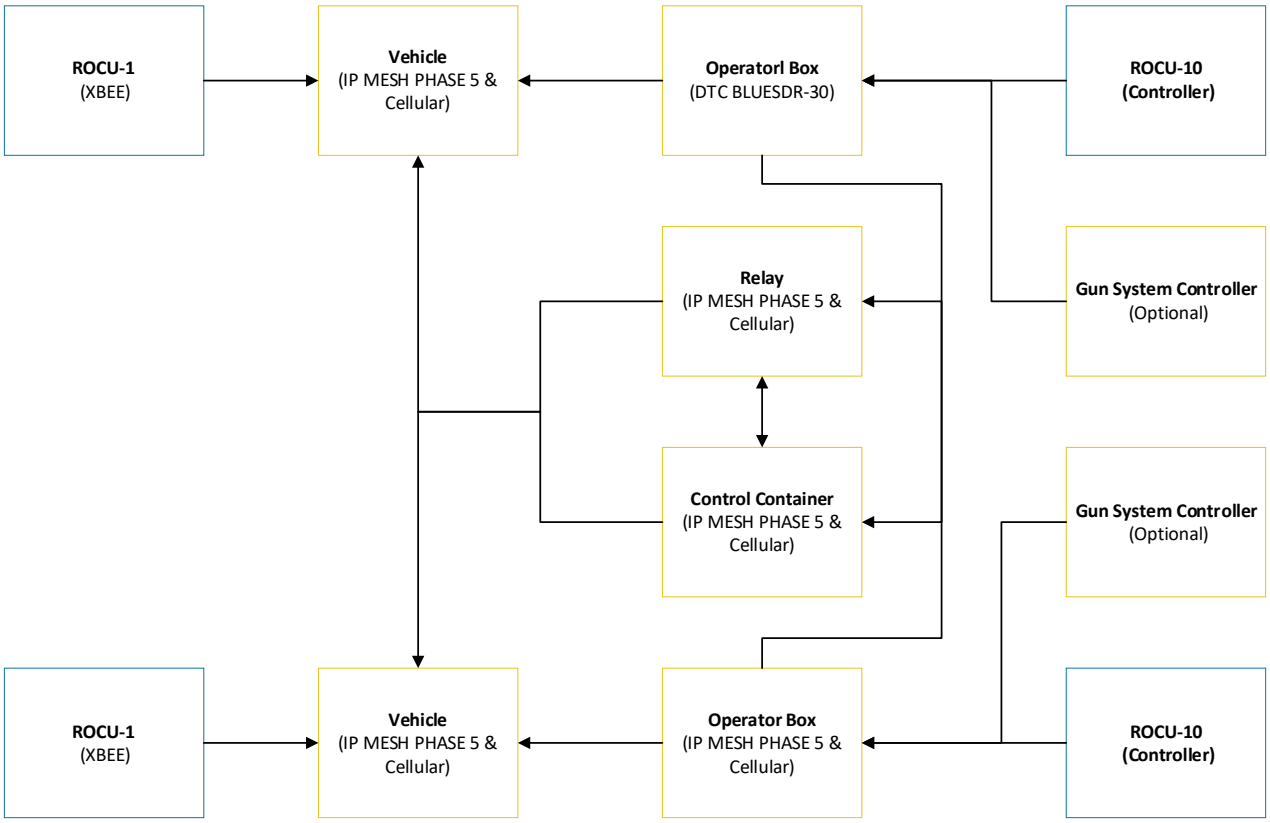


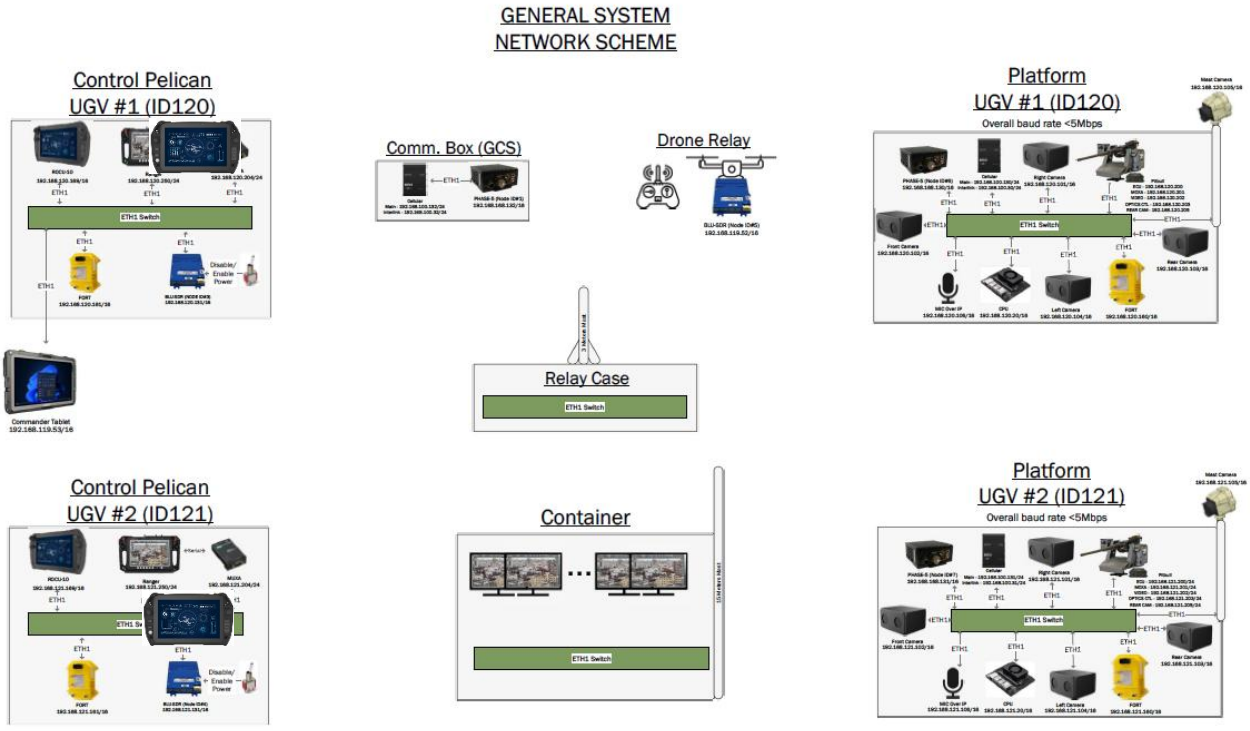
- Technical Specification

Parameter	Description
Mechanical/Environmental	
Radome ingress protection rating	IP65
Radome material	UV-resistant plastic and cable
Housing dimensions	Ø 110.0 x 53.0 mm
Cable	5x RG174, 3000 mm
Operating temperature	-40 ~ +85 °C
Weight	525 ± 10 g
UV resistance	V
RoHS Compliant	V

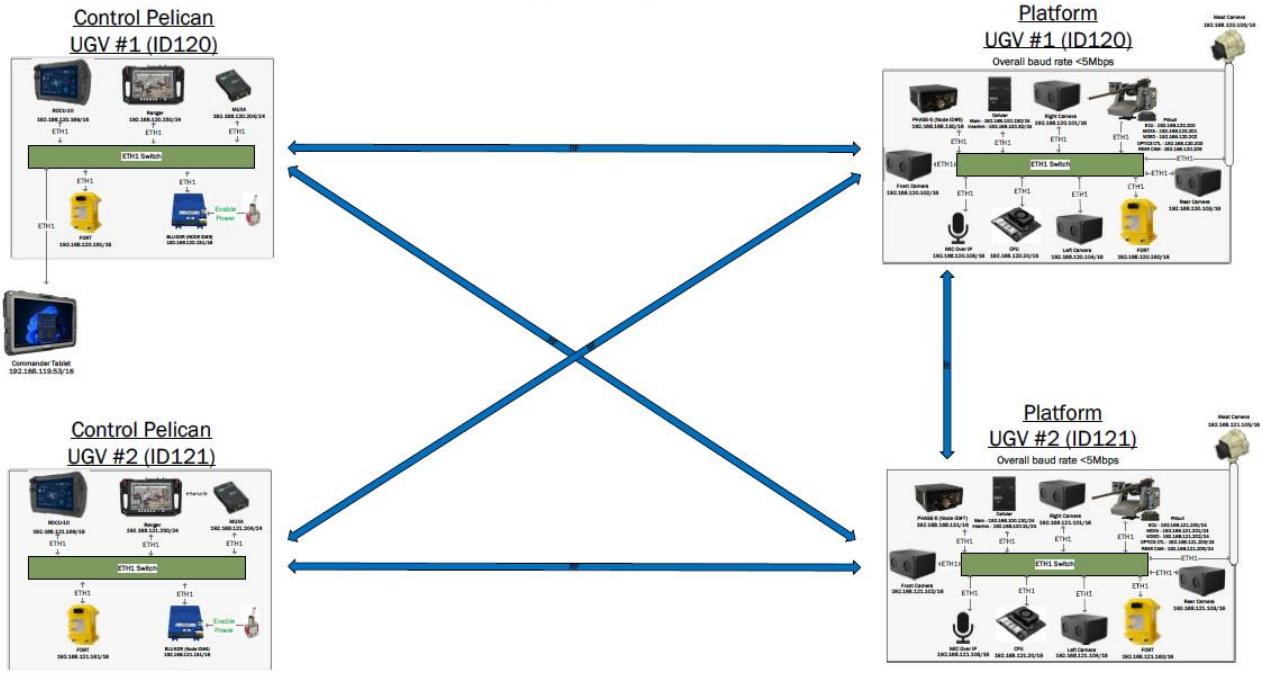
LTE MIMO ANTENNA	
Frequency range	698 - 960, 1710 - 2690 MHz
Polarization	Linear
Gain	3.0 dBi
V.S.W.R	<3.0
Impedance	50 Ω
Connector type and count	4x SMA Male
GNSS ANTENNA	
Frequency range	1575.42 ± 3 MHz
Polarization	RHCP
LNA gain	28 ± 2 dB
LNA noise figure	≤1.5 dB
V.S.W.R	<2.0
Impedance	50 Ω
Supply voltage	3 - 5 VDC
Current consumption	<15 mA
Connector type and count	SMA Male

3.6.4. System Communication Layout

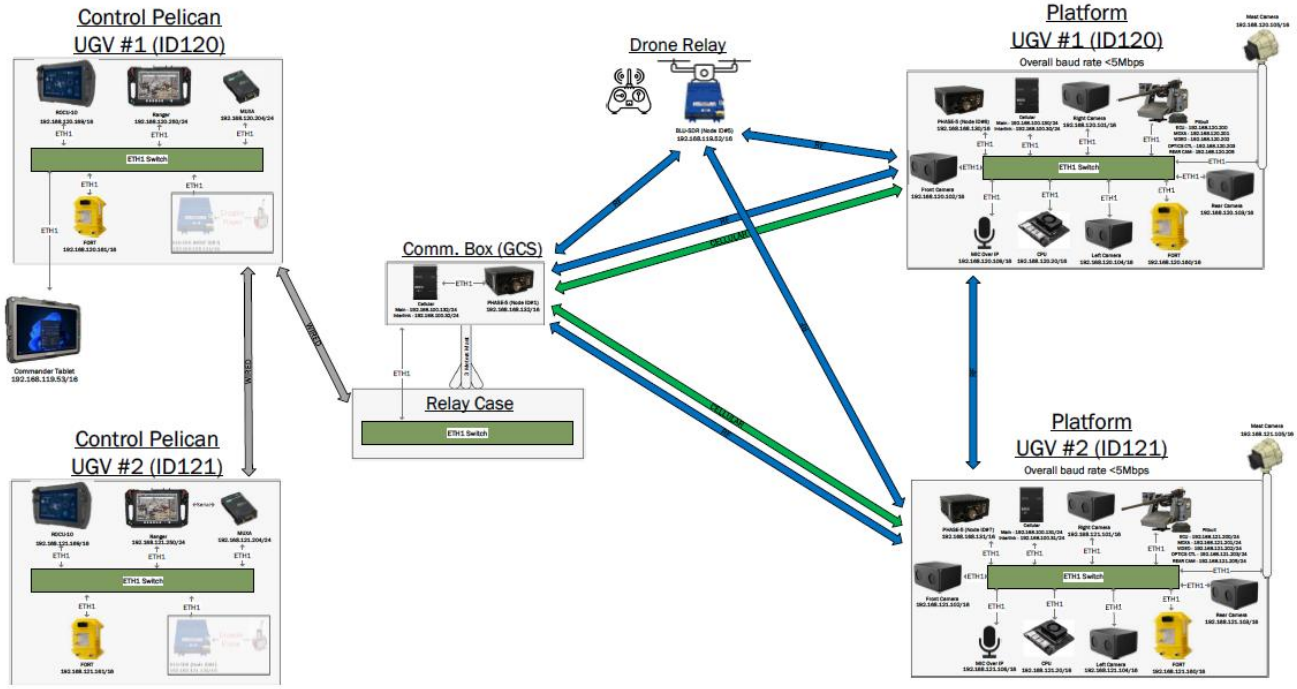


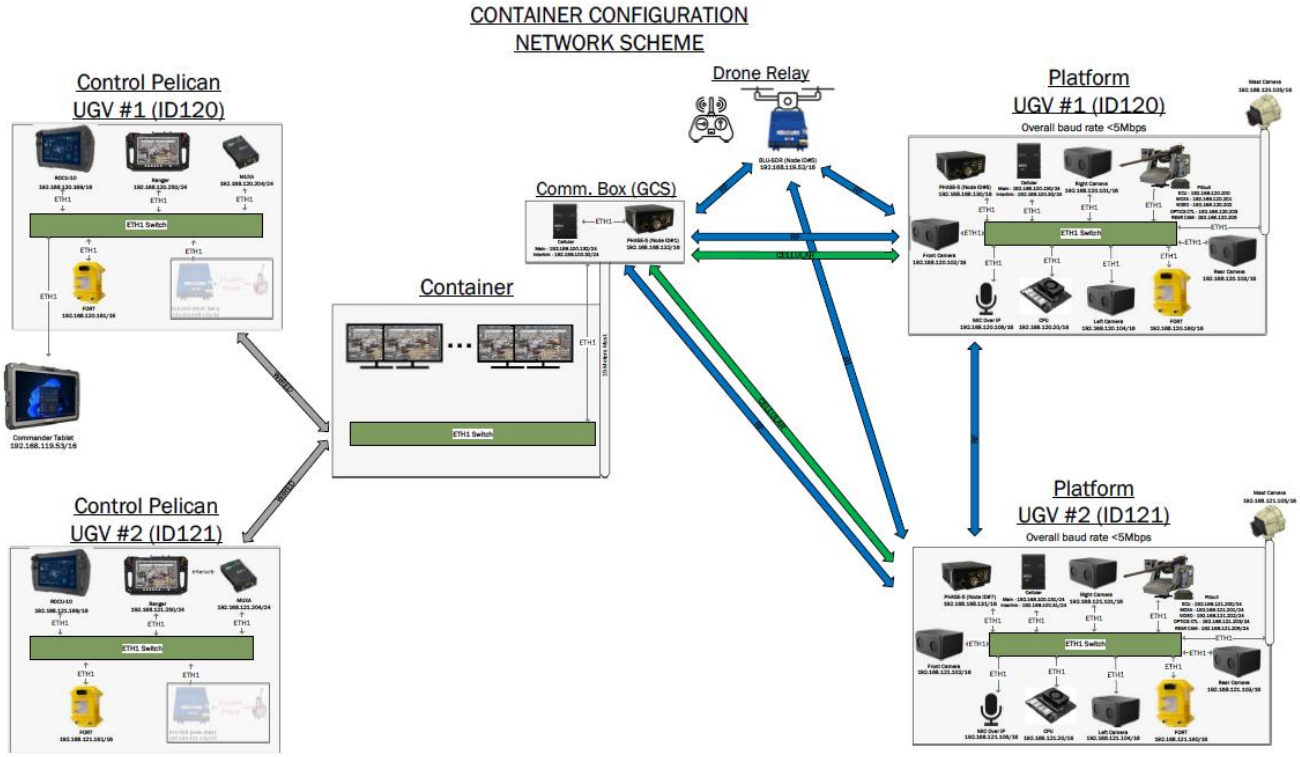


TACTICAL (BLU-SDR) CONFIGURATION NETWORK SCHEME



RELAY CASE CONFIGURATION NETWORK SCHEME



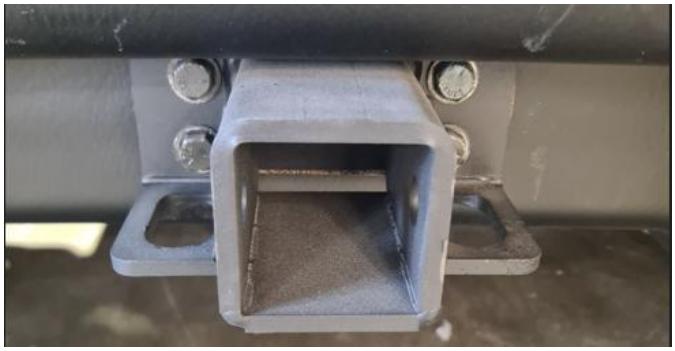

CONTAINER CONFIGURATION
NETWORK SCHEME
Control Pelican
UGV #1 (ID120)
Drone Relay
Platform
UGV #1 (ID120)
Overall baud rate <5Mbps
Comm. Box (GCS)
Container
Control Pelican
UGV #2 (ID121)
Platform
UGV #2 (ID121)
Overall baud rate <5Mbps
ETH1 Switch

3.7. Mechanical Interfaces

3.7.1. Overview

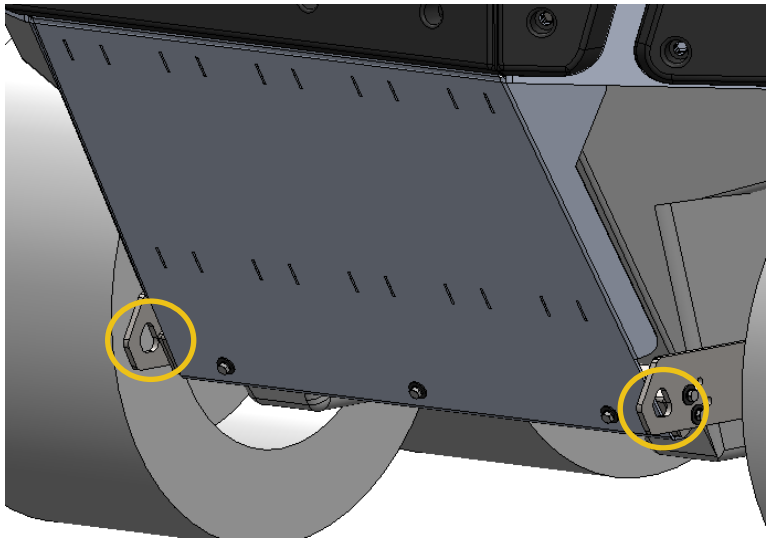
- Vehicle
  - Body (Back)

Towing Capacity: 1800LB (816KG)

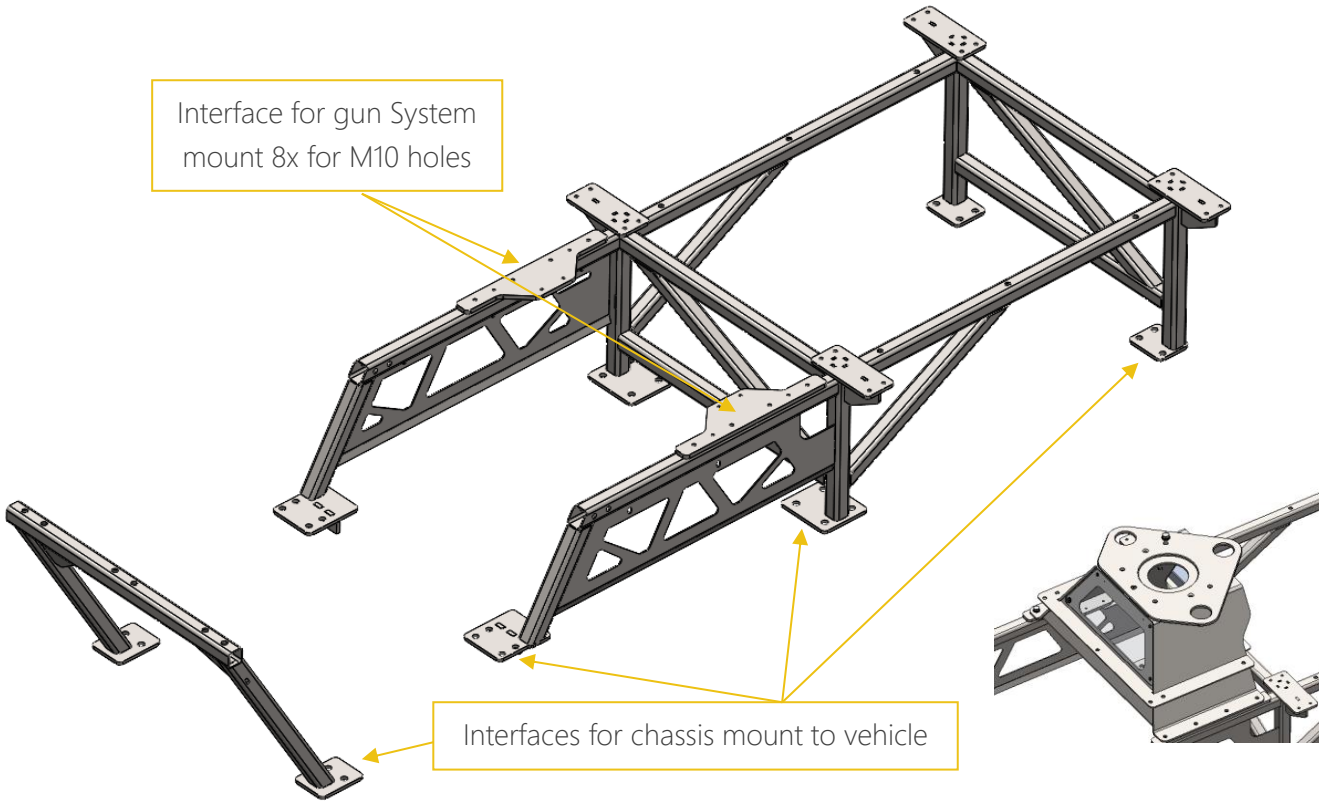


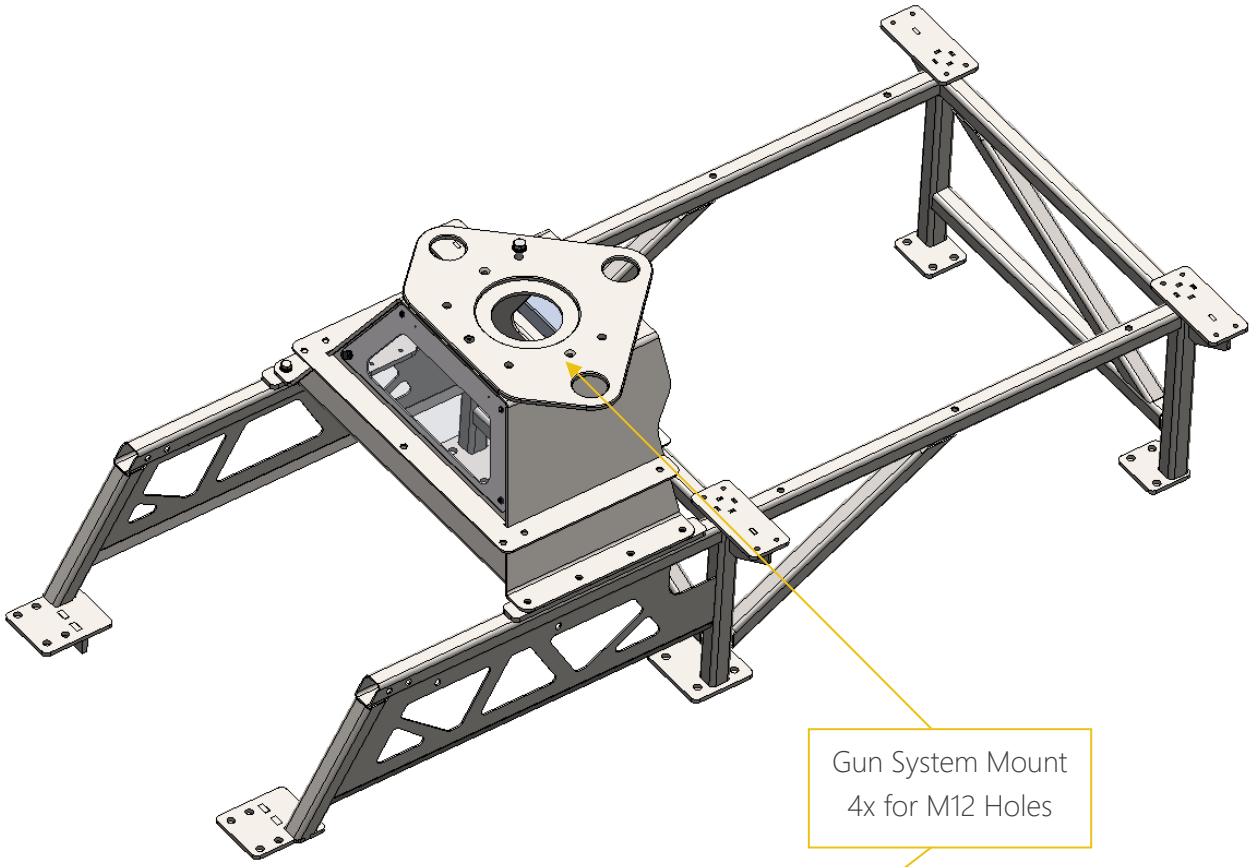
  - Body (Front)

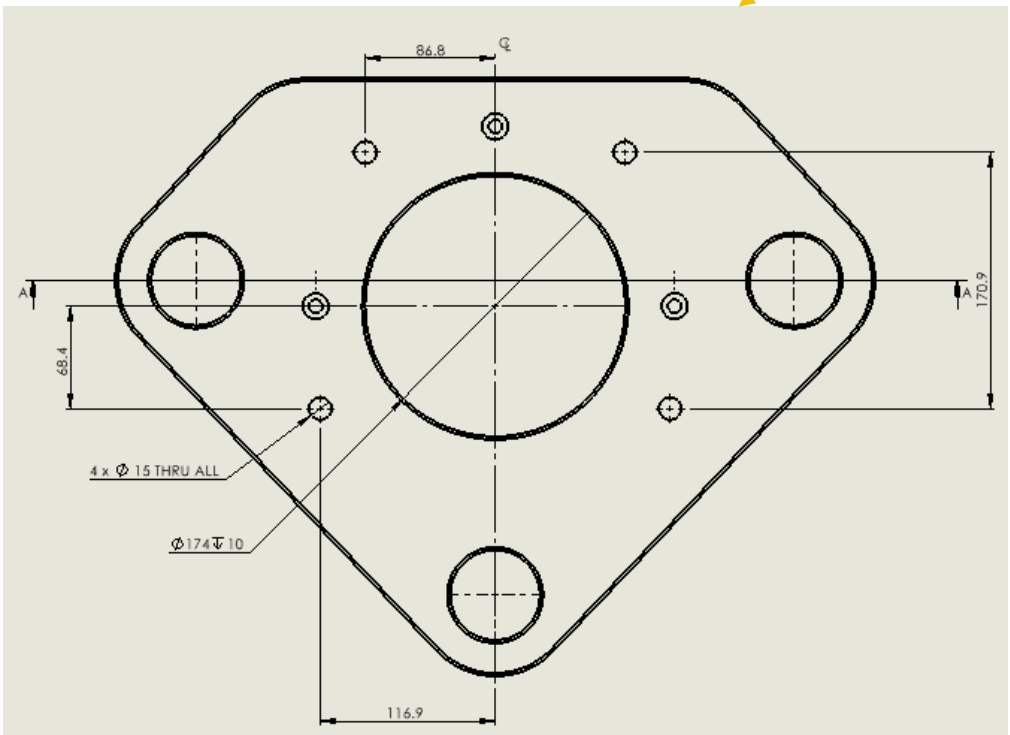
Towing Capacity: 1800LB (816KG)



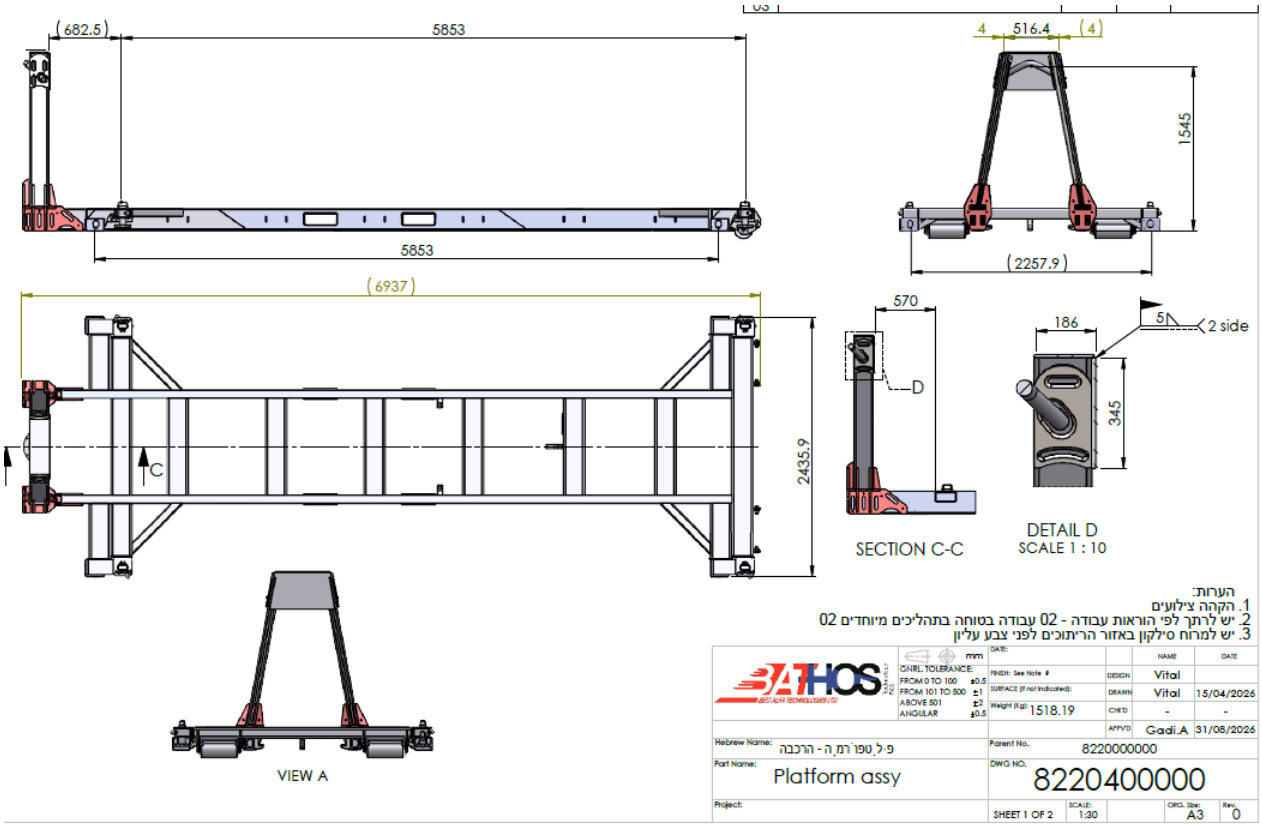
- Main Structure (Main Chassis)




Gun System Mount
4x for M12 Holes


86.8
A
68.4
170.9
4 x Ø 15 THRU ALL
Ø174↧10
116.9

- Container (RAMSA for transportation)
  The RAMSA is welded to the container .



3.8. Hardware Interfaces

3.8.1. Vehicle

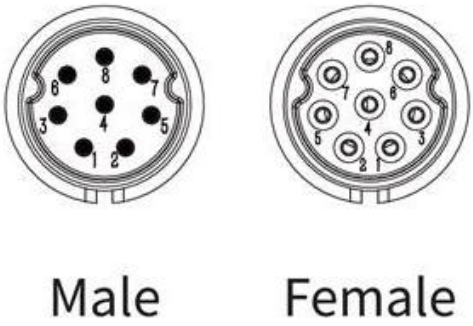
- Control Panel

Technician Interface:

Receptacle – CON01078 CNLINKO LP-16-J08SX-03-401

Plug – CON01079 CNLINKO LP-16-C08PE-01-001

Pin	Description
1	ETH_TX+
2	ETH_TX-
3	ETH_RX+
4	GND
5	CAN_L
6	ETH_RX-
7	CAN_H
8	N.C.

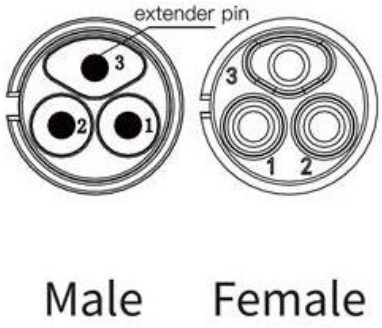


Charger Interface:

Receptacle – CON01064 CNLINKO LP-24-J03SX-03-435

Plug – CON01065 CNLINKO LP-24-C03PE-01-035

Pin	Description
1	12V_CHARGE_IN
2	N.C.
3	GND



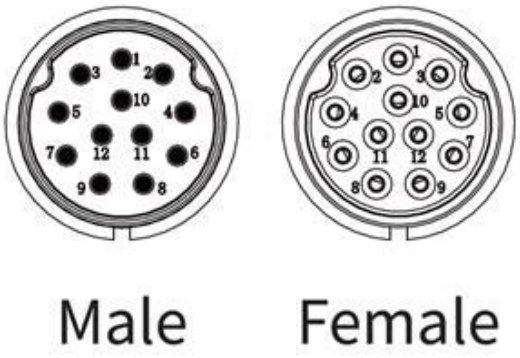
- Power Box (Mount to main structure under covers)

12V & Ethernet Interface:

Receptacle – CON01045 CNLINKO LP-20-J12SX-03-401A

Plug – CON01046 CNLINKO LP-20-C12PE-01-001A

Pin	Description
1	ETH_D0+
2	ETH_D0 -
3	ETH_D1+
4	ETH_D2+
5	ETH_D2-
6	ETH_D1-
7	ETH_D3+
8	ETH_D3-
9	12V_OUT
10	GND
11	GND
12	12V_OUT



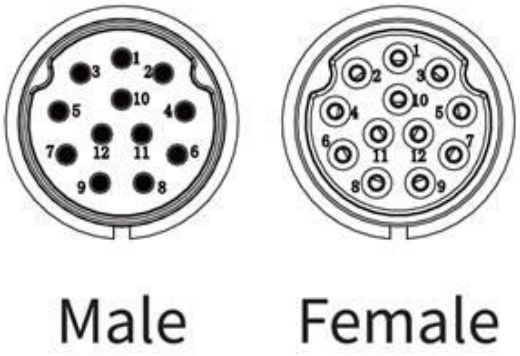
3.8.2. Operator Box

12V & Ethernet Interface:

Receptacle – CON01045 CNLINKO LP-20-J12SX-03-401A

Plug – CON01046 CNLINKO LP-20-C12PE-01-001A

Pin	Description
1	ETH_D0+
2	ETH_D0 -
3	ETH_D1+
4	ETH_D2+
5	ETH_D2-
6	ETH_D1-
7	ETH_D3+
8	ETH_D3-
9	12V_OUT
10	GND
11	GND
12	12V_OUT

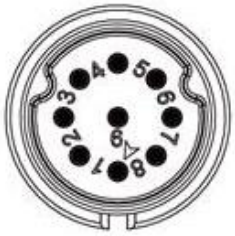


Ethernet Interface:

Receptacle – CON01039 CNLINKO LP-16-J09SX-03-401

Plug – CON01040 CNLINKO LP-16-C09PE-01-001

Pin	Description
1	ETH_D0+
2	ETH_D0 -
3	ETH_D1+
4	ETH_D2+
5	ETH_D2-
6	ETH_D1-
7	ETH_D3+
8	ETH_D3-
9	N.C.



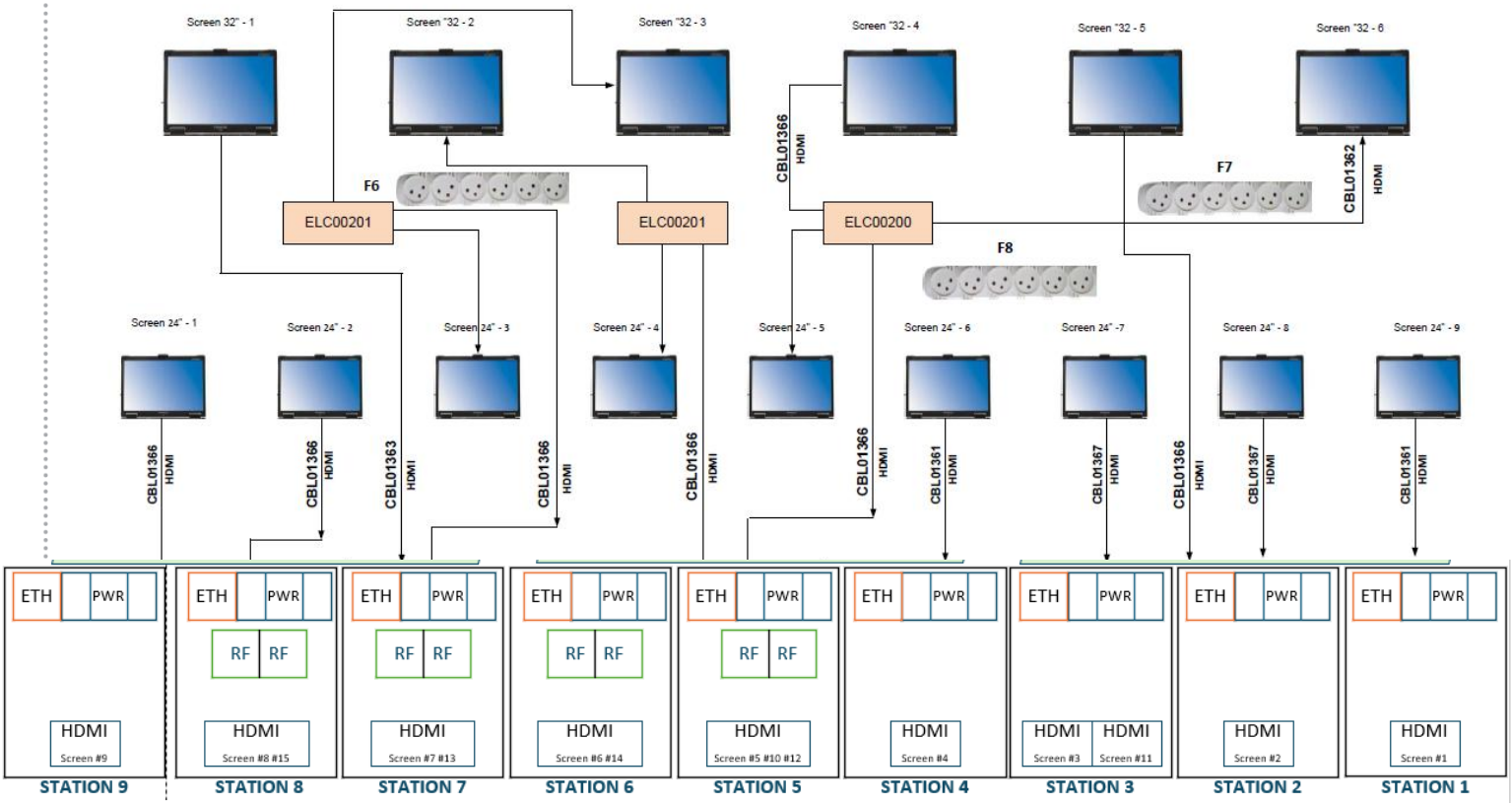
Male



Female

3.8.3. Tactical Operations Center (Controlled Container)

Nine operation stations, each including three power sockets, one HDMI socket, and one ETH socket.
Four stations have an additional two RF sockets (for RF antennas).



SMA



HDMI



RJ45



230 V



3.9. Gun System

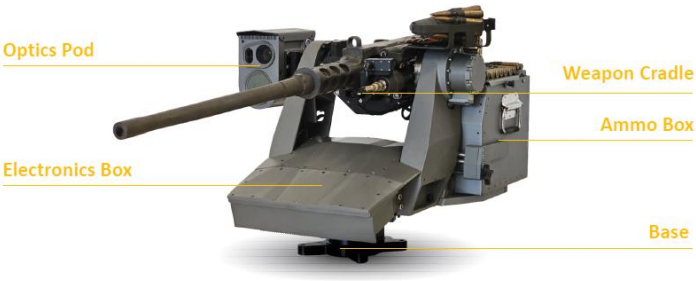
3.9.1. Overview

Pitbull gun system by General Robotics.





- High-definition (HD) video camera
- Thermal infra-red(IR) video camera
- Integrated eye-safe laser range finder (ESLRF)
- Rear compact optics for gun view
- Powerful optical tracker – sightline SLA
- "Point & Shoot™" technology
- Fire-inhibited zone algorithm (FIZ)
- Anti-collision system algorithm (ACS)
- Target prediction algorithm (TPA)
- Anti-drone capabilities
- Safe wire or wireless communication
- Multi-user & Multi-system (MUMS)
- Solenoid Safety pin and trigger – normally open
- Safe monitoring for live view





3.9.2. Technical specifications

- Gun System

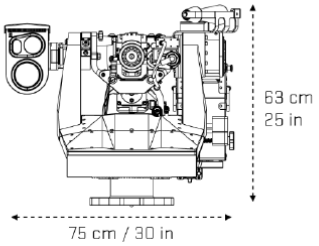
Parameter	Description
Dimensions w/o weapon	75x175x63 cm / 30x69x25 in (L x W x H)
Rotation Diameter with weapon	193 cm / 76 in
Weight (w/o weapon and ammunition)	< 95 kg / 209 lbs
Overall Combat Weight	< 165kg / 363 lbs
Mount type	Fast Mounting (4 bolts)
Main Armament	FN MAG Weapon Remote Cocking
Ammunition cartridge	200 rounds
Range Elevation	-30^0 to $+60^0$ Depending on platforms base
Operating Voltage	24 V – nominal (22-36V) supplied from the vehicle
Power consumption	80W average consumption
Traverse Range	$N*360^0$
Accuracy	0.4 Milliradian
Traverse and Tilt Speed	90^0 Per Second
Optics module	Uncooled Thermal camera: Continuous Optical Zoom 26-105 mm / 0.9-4.1 in Human target detection range < 1000 m / 0.62 miles Color HD Day camera: Continuous optical zoom X 40 Human target detection range < 1600 m / 0.99 miles Point & Shoot Technology Day & Night Target Tracking Automatic Target Detection
Dual Axis stabilization	0.01 degrees accuracy low drift (0.01 milliradian per minute)
Laser Range Finder	5.1 km / 3.1 miles (10 km / 6.2 miles with laser pointer option) Eye Safe Class 1- 1550 nm wave length

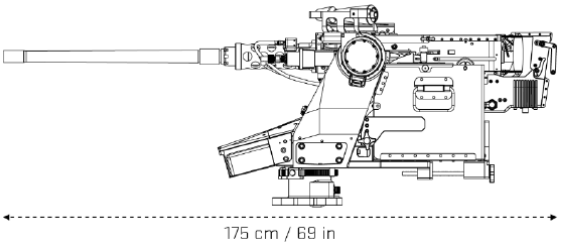
Environmental	MIL-STD-810G, MIL-STD-882E, MIL-STD-461F, MIL-STD-464, MIL-STD-1275D, MIL-STD-781D, MILSTD-2165

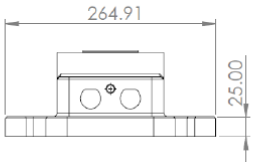
- Controller

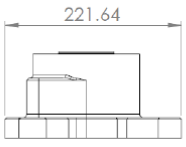
Parameter	Description
Dimensions	36 x 6 x 21 (L x W x H) CM
Weight	2.8 KG
Operating voltage	12-36 VDC (supplied from the C.B)
Operation Temperature	-30°to +70°C
Features	Point & shoot technology™ FIZ & ACS Safety interface Display real-time video camera of weapon and ammunition status Ammunition counter
Sensors	Up to 8 integrated sensors for automatic transition to safety mode. These sensors are used for the FIZ and ACS algorithms that installed in the system
Environmental	Design to Mil-std-810G, IP65

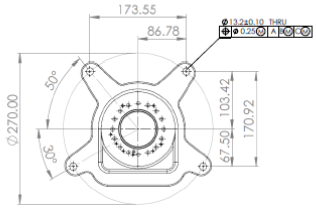
3.9.2.1. Mechanical Interface

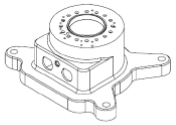






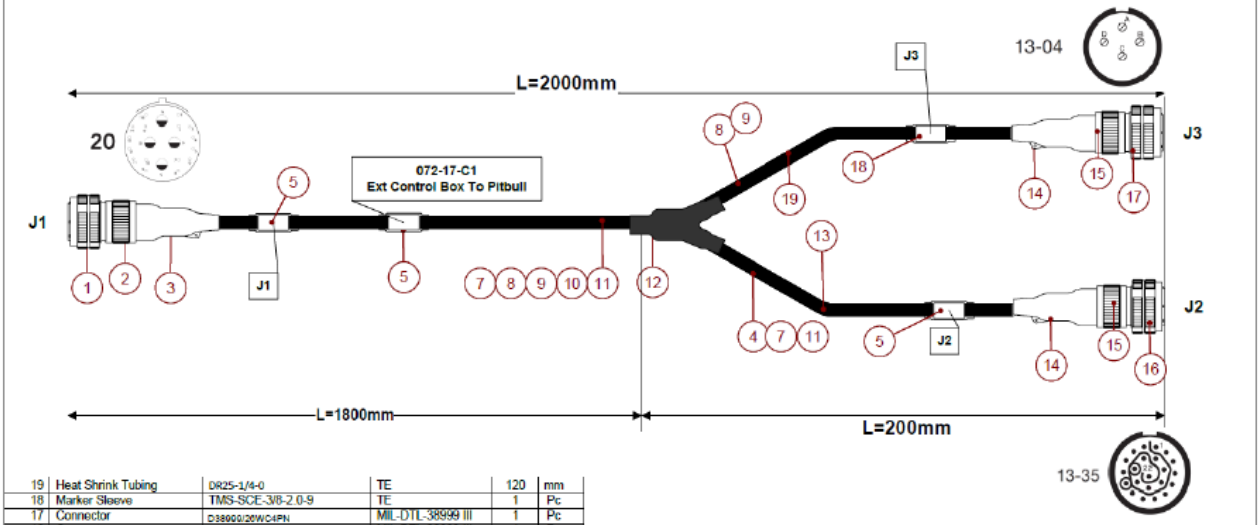






3.9.2.2. HW Interface

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
B0	Modified	11/4/21	
C1	P3.3	12/7/23	



19	Heat Shrink Tubing	DR25-1/4-0	TE	120	mm
18	Marker Sleeve	TMS-SCE-3/8-2.0-9	TE	1	Pc
17	Connector	D38999/26WC4PN	MIL-DTL-38999 III	1	Pc
16	Connector	D38999/26WC35PN	MIL-DTL-38999 III	1	Pc
15	Adaptor	M85049/88-13W03	MIL-DTL-38999	2	Pc
14	Boot	202K121-25-0	TE	2	Pc
13	Heat Shrink Tubing	DR25-3/8-0	TE	120	mm
12	MOLDED BOOT	382A012-25-0	TE	1	Pc
11	Wire	55A0111-24-9	TE	280	mm
10	Heat Shrink Tubing	DR25-1/2-0	TE	160	mm
9	Wire	55A0111-16-2	TE	560	mm
8	Wire	55A0111-16-0	TE	560	mm
7	Cat5e	E50824	PIC	280	mm
5	Marker Sleeve	TMS-SCE-1/2-2.0-9	TE	3	Pc
4	Wire	M27500024SB2D00	TE	6000	Pc
3	Boot	202K132-25-0	TE	1	Pc
2	Adaptor	M85049/88-17W03	MIL-DTL-38999	1	Pc
1	Connector	80S17W20PN	MIL-DTL-38999 III	1	Pc

Notes:
A) Fill all empty crimps in connectors for sealing.

Tolerance Table				
Dimension (mm)	Tolerance (mm)		NAME	DATE
0-300	-0 + 10	DESIGN	Yuval	07/12/2023
300-500	-0 + 50	DRAWN	Yuval	07/12/2023
1500-3000	-0 + 100	APVD	Inon	07/12/2023
Over 3000	-0 + 5%	Security: U.C		

GENERAL ROBOTICS
Fighters' Best Partners
www.grobotics.com

Title: 072-017
Description: Ext Control Box To Pitbull
All dimensions in millimeters | Sheet 1 Of 2

